Ethical Extensions to a Federated Learning Framework for Single-Cell RNA Sequencing

Ethics in Artificial Intelligence Project - 6 CFU

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Abstract

Federated Learning (FL) enables multiple institutions to collaboratively train Machine Learning models without exchanging sensitive data, making it particularly well suited for biomedical applications, where patients may be reluctant to share their personal clinical records with external parties involved in AI development. In this project, we build upon an existing FL framework based on Flower and scVI, developed by the Swiss Institute of Bioinformatics (SIB) and described by Malpetti et al. [4] for the collaborative analysis of single-cell RNA sequencing data. The framework demonstrates how multiple institutions can jointly train a model while keeping patient data local, but it does not explicitly investigate important ethical concerns associated with Federated Learning. Thus, we propose two complementary extensions addressing privacy and fairness, respectively. The first investigates whether model updates exchanged during training may leak information about the institution that generated them through a Source Identification Attack and about properties of each client's local dataset via a series of Property Inference Attacks. We then evaluate Differential Privacy as a possible mitigation strategy, by quantifying the trade-off between privacy protection, measured through reduced attack effectiveness, and model utility. The second extension investigates whether the standard FedAvg aggregation strategy disproportionately favors institutions with larger local datasets and evaluates alternative aggregation methods aimed at improving fairness across clients. It further examines the effect of partial client participation on model performance and fairness, thereby simulating more realistic Federated Learning deployments in which not all institutions may participate in every training round.

Acknowledgements

The work by Malpetti et al. [6] served as the primary reference and theoretical foundation for this project. The original Federated Learning framework on which this work is based was developed by the authors at the Swiss Institute of Bioinformatics (SIB). We would like to express our sincere gratitude to Daniele Malpetti for granting us access to the original framework and for giving us the opportunity to extend the project with the ethical analysis presented in this report.

Contents

1 Introduction to Federated Learning

Federated learning (FL) is a collaborative Machine Learning paradigm that allows multiple devices or institutions to jointly train a shared model without directly exchanging their local data with third parties. Unlike traditional centralized approaches, FL allows information to remain under the control of the institution that collected or generated them, potentially strengthening privacy protection. This makes FL particularly suitable for biomedical research, where patient data are highly sensitive and subject to strict sharing regulations. Instead of sharing individual records, participating institutions exchange model updates that capture the information learned from their local datasets.

In a typical FL setting, data holders, commonly referred to as clients, form a consortium coordinated by a central server. The server initializes a global model and distributes its parameters to the participating clients. Each client uses these parameters to initialize a local model, which is then trained on its local data. The resulting local model parameters or updates are sent back to the server, which aggregates them to produce an updated global model. The updated parameters are subsequently redistributed to the clients and the process is repeated for multiple training rounds, until a predefined stopping criterion is met. An overview of this workflow is shown in Figure 1.

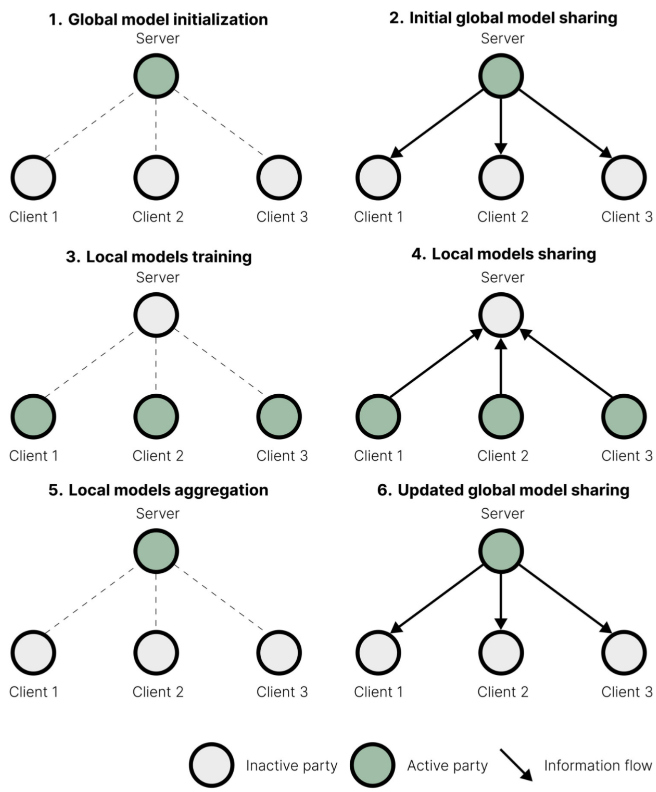

Figure 1: Federated Learning workflow

In each round, the central server therefore acts as the coordinator of the training process, while clients perform local optimization using data that remain within their respective institutions. Arrows in *Figure 1* indicate the direction of information flow within the consortium, while parties active at each step are represented in green. This iterative process allows the consortium to train a shared model without requiring direct access to the underlying dataset. The topology of a FL consortium is determined by the number of participating parties and how they interact. The most common configuration is a centralized topology, in which a central server, or aggregator, collects model updates from the clients, computes the global model and redistributes back the updated parameters. Clients do not communicate directly with one another but interact exclusively with the central server. This is the particular topology adopted in this work. In a decentralized topology, by contrast, there is no dedicated aggregation server: participants can act as both trainers and aggregators and exchange information directly through peer-to-peer communication. Hybrid topologies are also possible, in which elements of both approaches are combined; for example, multiple servers may communicate with one another and with their clients, while direct communication between clients remains restricted. These alternative configurations are illustrated in *Figure 2*.

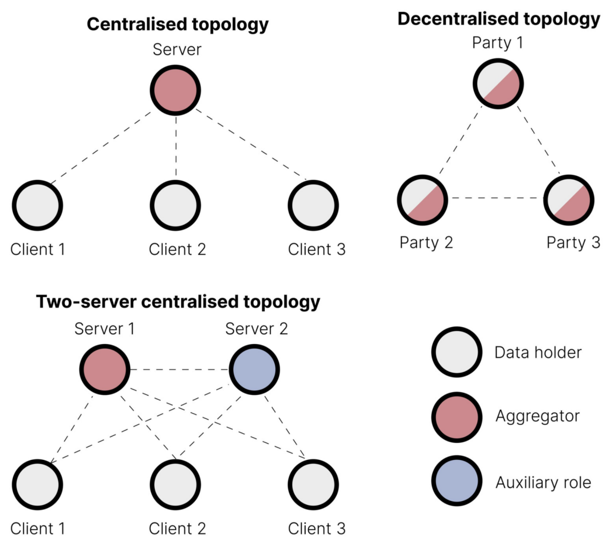

Figure 2: Alternative Federated Learning topologies

The terminology adopted to describe participants in the consortium can also depend on the selected topology. In centralized FL, data holders are typically referred to as clients, reflecting their interaction with a central server. In decentralized settings, where no single central entity coordinates the training process, participants are more generally referred to as parties. Beyond topology, FL settings can also be characterized by how data are partitioned across institutions. Two common configurations are horizontal and vertical Federated Learning. In horizontal FL, different parties hold data from different individuals but share the same feature space. For example, several institutions may each hold single-cell sequencing data from distinct patient cohorts, with the data generated and represented using the same features. In vertical FL, by contrast, different parties hold different features for the same individuals. For example, one institution may hold one omics layer while another holds a different omics layer or complementary information for the same subjects. Horizontal FL is the predominant setting in bioinformatics and is indeed the configuration considered in this work.

A further defining characteristic of FL, particularly in horizontal settings, is data heterogeneity. The amount of data available to each client, the distribution of features, patient populations, sample sizes, experimental protocols and data-generation technologies can vary substantially across institutions. This heterogeneity is commonly referred to as the non-IID nature of federated data. Although it introduces challenges for optimization and model aggregation, it is also one of the strongest motivations for FL in biomedical applications: combining data from heterogeneous institutions can expose the global model to a broader range of biological variability than would be available from any individual institution, potentially improving its ability to generalize across sites. This aspect is further investigated in Section 5 - Data Heterogeneity.

2 Federated Learning in Bioinformatics

The characteristics of Federated Learning outlined in the previous section make it especially well suited to healthcare and bioinformatics applications. In these domains, data are often distributed across institutions, which are subject to strict privacy and regulatory constraints. As previously mentioned, a typical FL consortium consists of two main components: clients, which hold local datasets and perform model training, and an aggregation server, which coordinates the overall learning process. In healthcare settings, clients may correspond to hospitals, research institutions or other data holders, each contributing to the training process with their local patient data, stored within their infrastructure. Several studies have reported improvements over single-institution analyses when applying FL to biomedical tasks. For example, FL has been shown to improve breast density classification, with reported increases of up to 6% in accuracy and 46% in generalizability. Similarly, FL-based approaches for COVID-19 outcome prediction have documented improvements of up to 16% and 38% for predictions at 24 and 72 hours, respectively. Applications to rare tumour segmentation have reported improvements of approximately 23-33% and 15% across the metrics considered. In drug discovery, a consortium of ten pharmaceutical companies demonstrated that FL enhanced quantitative structure-activity relationship (QSAR) models by 12% on both evaluated metrics. Early applications using electronic health records have also highlighted predictive performance comparable to that obtained when training on centrally pooled data, without substantial practical degradation in model performance. Beyond raw performance, broader adoption of FL could enable academic and clinical institutions to collaboratively exploit complementary sources of biomedical information, which may be underrepresented or unavailable within any single biobank.

However, the benefits of FL should not be interpreted as automatic or unconditional. Recent research has cautioned that some of the expected advantages of FL in healthcare may be exaggerated, misleading or strongly dependent on the specific experimental and organizational setting. In particular, the same decentralization that limits direct access to patient data can also reduce the transparency of the resulting learning process: stakeholders may have limited ability to inspect, analyze or curate the data distributions underlying the trained model. This lack of visibility raises important concerns regarding fairness, accountability and governance, and can also introduce additional vulnerabilities, including susceptibility to data poisoning attacks. In particular, this work focuses on single-cell RNA sequencing (scRNA-seq), a technology that specifically measures gene expression at the level of individual cells rather than averaging expression across an entire tissue, as in a standard RNA sequencing pipeline. This higher resolution makes it possible to characterize distinct cell populations within a tissue and to identify their specific gene-expression profiles. However, it also introduces substantial heterogeneity across datasets, as institutions differ in the cellular composition they store, sample sizes and sequencing technologies adopted. These characteristics make scRNA-seq a particularly relevant scenario for investigating both the privacy and fairness implications of Federated Learning.

3 Employed Technologies

We used the Flower (FLWR) framework to simulate a cross-silo federated learning environment with multiple clients and a central server. Flower allows each client to independently train a local model and periodically synchronize with the server using a specified aggregation strategy. Its modular design and ease of customization make it particularly useful for large-scale and multi-omics studies involving heterogeneous devices and clients.

The federated learning experiments were implemented using the Flower framework and its message-based interface. In this interface, each federated round is represented as an explicit exchange of messages between the server and the participating clients. This provides direct control over the different stages of the federated learning process while still allowing the experiments to build on Flower's standard `FedAvg` strategy. Specific components of the training process, such as client selection and model-update aggregation, could therefore be customized without having to reimplement the complete federated learning pipeline. All experiments were executed using Flower's local simulation mode. The simulation represents seven independent clients, each corresponding to one institution and its associated sequencing technology, communicating with a central server process on a single machine.

Each experiment was launched from the command line with its own configuration. The federative model was based on *scVI*, a variational autoencoder designed for single-cell RNA sequencing data. The model was implemented using the `scvi-tools` library and relies on PyTorch for the underlying deep-learning computations. scVI directly models gene expression count data, making it suitable for the single-cell datasets considered in this project. The datasets were represented as `AnnData` objects. Scanpy was used for standard single-cell data preprocessing and manipulation, while scikit-learn was used to partition the data available at each client into training and validation subsets. The resulting client-specific training and validation sets were then used throughout the federated experiments to train the shared model and evaluate its performance independently across clients. For analysis and reporting, we used standard Python tools throughout: NumPy and Pandas for handling numbers and tables, Matplotlib for all plots, and Jupyter notebooks both for early exploratory work and for comparing results across our different experiments.

4 Original Work - Swiss Institute of Bioinformatics

The original Federated Learning framework developed by the Swiss Institute of Bioinformatics (SIB) for the collaborative analysis of single-cell RNA sequencing data provides a simplified simulation of a realistic multi-institutional collaboration involving seven institutions worldwide. Each institution holds a pancreas single-cell dataset generated using a different sequencing technology: *CelSeq, CelSeq2, Fluidigm C1, Smartseq2, inDrop1, inDrop2, inDrop3, inDrop4 and SMARTer*. Institutions collaborate to train an open-source model designed to remove technology-related batch effects, using an architecture provided by scVI-tools, with the broader objective of disseminating the resulting model through a scientific publication. Since data cannot be shared across institutions, the consortium adopts a Federated Learning approach implemented using the Flower framework. Each institution performs training locally on its own dataset, while only model updates are exchanged with the central server. The framework also considers a downstream use case, involving a separate institution that holds its own pancreatic single-cell dataset and would accesses the trained model after its publication. This institution combines measurements generated using two different sequencing technologies, because one sequencing machine failed during the study and was thus subsequently replaced. As a result, the dataset exhibits the same type of technology-related batch effects that the federated model was designed to correct. The original framework, therefore, demonstrates both how to develop a collaborative training approach across institutions and a real-world scenario, in which such model is subsequently applied to data from a new, previously unseen institution.

The baseline framework provides four main functionalities:

- **Federated identification of highly variable genes:** the framework runs the highly variable gene selection procedure in both a centralized and federated settings and demonstrates that the two approaches produce identical results;
- **Centralized and federated scVI training:** a Deep Learning model based on scVI is trained using both centralized and Federated Learning approaches;
- **Evaluation on unseen data:** the effectiveness of the federated and centralized models is compared on data originating from a new institution that did not participate in the training process;
- **Live federated training:** the framework supports an interactive federated training session in which participants can form small groups, with one participant acting as the server and the others as clients.

4.1 Federated Highly Variable Genes (HVG)

This first exercise identifies the top-k most highly variable genes (HVGs) in a federated dataset. Following the reproducibility steps described in the last section of this report, the dataset is downloaded and organized into two separate directory structures: one for the centralized and one for the federated setting. The centralized data directory contains the datasets used for model training, consisting of separate training and validation sets, together with two metadata files: the ordered list of all genes and the list of sequencing technologies represented in the dataset. It also contains the external test set subsequently used to compare the performance of the centralized and federated models. The federated data directory, in contrast, is organized into one subdirectory for each client and a separate directory for the server. Each client directory contains the corresponding local portions of the training and validation datasets, together with the associated metadata files. The server directory contains only the metadata and does not contain any expression data. This organization reflects the intended federated setting: each client retains exclusive access to its local data, while the server coordinates the computation without directly accessing the underlying expression data.

Within the `exercise-hvg` folder, `centralized_analysis.ipynb` implements two helper functions that decompose the built-in HVG routine into separate steps. This illustrates how a standard centralized computation can be reformulated in an additive form, which is essential in a federated setting because it allows the global result to be reconstructed without pooling the underlying data. The folder also contains `pyproject.toml`, which configures the Flower application, and `app/client_app.py`, which defines the client-side Flower application. In its initial, minimal form, each client sends a single random local value to the server during training, together with an associated weight of one; no actual model is trained at this stage. The server aggregates these values using FedAvg with equal weights, providing a simple working skeleton that is subsequently adapted to the actual HVG computation. A first simple evaluation endpoint is also defined in the client application, to verify communication from the server back to the clients. With the evaluation fraction on the server set to 1.0, every client receives the average of the random integers computed by the server and prints the result locally. Since the exercise is executed in simulation mode, the outputs from all clients appear in the same terminal; in a real deployment, each client would instead receive and print its result within its own local environment. Computing the HVGs requires three quantities: the gene-wise sums of expression values, the gene-wise sums of squared expression values and the total number of cells. Each client reads its local data, computes these statistics using a custom helper function and sends the resulting values to the server. On the server side, a custom strategy defined in `custom_strategy.py`, which inherits from FedAvg, reconstructs the global sums. These global statistics are then passed to the same helper function, to compute the corresponding global statistics and identify the highly variable genes. Because communication between the Flower server and clients is restricted to messages containing NumPy arrays, the server cannot directly return gene names. Instead, it constructs a binary mask with one entry per gene, where a value of 1 indicates that the corresponding gene has been selected as highly variable and 0 otherwise. Since the server and all clients share the same ordered list of genes, each client can apply the mask locally, to recover the corresponding gene names during evaluation. Finally, `results_comparison.ipynb` verifies that the centralized and federated HVG lists are identical. The comparison is shown explicitly for client 0, but the same result holds for all clients because every client receives the same information from the server.

4.2 Federated scVI: Training, Monitoring and Model Release

The second exercise trains a scVI model in a federated simulation using Flower, saves the final aggregated model, monitors training and validation losses across communication rounds, introduces early stopping and compares the resulting federated model with a centralized baseline. Within the `exercise-scvi` directory, `centralized_training.ipynb` trains the scVI model on the centralized training set and saves the resulting model as a baseline for comparison. The `app` subdirectory contains the following components:

- `server_app.py`: defines the server-side workflow. Since the server does not have access to the underlying data, it creates a dummy `AnnData` object containing the highly variable genes and batch categories. This object is used only to initialize the scVI model architecture. The resulting initial model weights are then passed to Flower as the starting point for federated training;
- `client_app.py`: defines the client-side training and evaluation logic. Each client loads its local training and validation data, restricts them to the selected highly variable genes, initializes a scVI model, receives the current global model weights, performs local training and returns the updated weights to the server;

- `custom_strategy.py`: contains several custom strategies of increasing complexity that extend Flower's FedAvg. These strategies implement functionalities for saving the final aggregated model, collecting and logging training and validation losses, generating training curves and performing early stopping;

- `task.py`: contains helper functions shared by the server and clients, including data loading, creation of dummy and local `AnnData` objects, scVI model initialization, extraction and loading of model weights, loss computation, configuration parsing and saving of the final trained model.

At the end of the training phase, the final aggregated model is saved on the server side as a single global model. This mirrors the intended use case in which the consortium releases one trained model for subsequent applications. The `FedAvgSaveModelPlotLosses` strategy saves this final model, collects training and validation losses from the participating clients, computes their global weighted averages and updates the corresponding loss plot after each communication round. To investigate the effect of the balance between local computation and communication, the total amount of local training is kept approximately fixed at 50 epochs while varying how this training budget is distributed across local epochs and federated communication rounds. For example, experiments are performed using 1 local epoch over 50 federated rounds, 5 local epochs over 10 rounds and 10 local epochs over 5 rounds. This allows the effect of different communication frequencies to be assessed while keeping the overall amount of local optimization comparable. Finally, `model_comparison.ipynb` loads the centralized and federated models, computes their latent representations and visualizes them using UMAP. The resulting representations show that the two models achieve comparable levels of technology-related batch-effect correction, indicating that the federated training procedure can reproduce the main biological representation learned by the centralized baseline, with the key advantage of keeping the training data local. The overall structure of the original project is summarized in Figure 3.

4.3 Live Federated Run

The final component of the original work consists of a live, multi-client federated training session, in which one participant acts as the server and the remaining participants act as clients. All participants connect through the same Wi-Fi network provided by a local router. In a real-world federated deployment, the server and clients would typically operate across separate networks and institutional infrastructures, but this setup is simplified here to make the exercise easier to reproduce. Nevertheless, the training process remains genuinely federated: the underlying data remain stored on individual laptops and only model updates are exchanged between the clients and the server during the communication. For the purposes of our Ethics in Artificial Intelligence project, this final experiment is reported for completeness but was not reproduced. The required local-network setup was not available to us and the live demonstration falls outside the scope of the ethical extensions investigated in this work, which instead focuses on the privacy and fairness implications associated with the federated training pipeline.

```
AI_ETHICS_Proj_Foschi_Triulzi-main/
├── FL
│   ├── exercise-hvg/
│   │   ├── centralized_hvg.ipynb
│   │   ├── results_comparison.ipynb
│   │   ├── pyproject.toml
│   │   └── app/
│   │       ├── client_app.py
│   │       ├── custom_strategy.py
│   │       ├── server_app.py
│   │       └── task.py
│   │
│   ├── data_centralized/
│   │   ├── all_genes.json
│   │   ├── all_techs.json
│   │   ├── top2k_genes.json
│   │   ├── pancreas_train.h5ad
│   │   ├── pancreas_valid.h5ad
│   │   └── pancreas_test.h5ad
│   │
│   ├── data_federated/
│   │   ├── data_client_0/
│   │       ├── all_genes.json
│   │       ├── all_techs.json
│   │       ├── top2k_genes.json
│   │       ├── train.h5ad
│   │       ├── valid.h5ad
│   │   ├── data_client_1 ... data_client_6/
│   │   └── data_server/
│   │       ├── all_genes.json
│   │       ├── all_techs.json
│   │       └── top2k_genes.json
│   │
│   |── model_centralized/
│   │   └── model.pt

│   │
│   |── model_federated/
│   │   ├── adata.h5ad
│   │   ├── global_loss.npy
│   │   └── model.pt
│   │
│   ├── exercise-scvi/
│   │   ├── centralized_training.ipynb
│   │   ├── model_comparison.ipynb
│   │   ├── pyproject.toml
│   │   └── app/
│   │       ├── server_app.py
│   │       ├── client_app.py
│   │       ├── custom_strategy.py
│   │       └── task.py
│   │

│   ├── preliminaries.py
│   ├── fl-course-env.yaml
│
└── README.md
```

Figure 3: Overview of the original project developed at the Swiss Institute of Bioinformatics (SIB)

5 Data Heterogeneity

In principle, the objective of FL is to minimize the privacy risks associated with data sharing while ensuring compliance with regulatory requirements above. However, variations in measurement protocols, patient populations and data distributions across parties make data harmonization necessary. This requirement becomes considerably challenging when direct access to the underlying data is restricted by design. The resulting difficulty is commonly referred to as a data heterogeneity problem. Federated datasets are typically non-identically and independently distributed (non-IID), i.e. training examples held by different clients may originate from substantially different distributions. In a federated genomic network, such heterogeneity arises naturally because training data originate from hospitals with different patient cohorts, databases, technical infrastructures and clinical equipment. Although techniques such as batch correction can mitigate such effects, the attempt to harmonize data may themselves introduce a trade-off with model performance.

Critically, data heterogeneity does not only affect model robustness and generalizability, but raises fairness concerns having direct implications also for privacy. When local datasets differ substantially in size or distribution, their corresponding updates may contribute unevenly to the aggregated model, potentially resulting in a global model that performs better for some institutions than for others. Consequently, despite contributing to the same federation, some hospitals and, by extension, their patients, may benefit less from the resulting system. At the same time, the statistical distinctiveness of a client's data distribution can itself constitute a privacy risk. If local updates retain information about characteristics that distinguish one client's dataset from those of other participants, an adversary may potentially exploit these signals to infer properties about that client's data or, in some cases, to identify its contribution to the federation. Data heterogeneity therefore represents a common structural challenge underlying both fairness and privacy: the same differences in data distribution that make a client's contribution less similar to those of other participants may also make its updates easier to distinguish from others. In this sense, the privacy and fairness concerns examined separately in this work can be understood as two manifestations of the same underlying structural property of federated systems.

6 Privacy Risks and Mitigation in Federated Learning

6.1 Motivation and Research Question: a Gap between Federated Learning Privacy Promise and Practice

Federated Learning has largely entered the medical Machine Learning landscape in response to a structural challenge: patient data are subject to increasingly stringent regulatory requirements, including the GDPR and the AI Act in the European Union and the HIPAA in the United States. These governing constraints limit the use and sharing of sensitive health data, making multi-centre studies and cross-institutional collaborations more difficult, potentially restricting the scale and diversity of datasets available for clinical research. FL is often presented as a technical response to this challenge, because it enables multiple institutions to collaboratively train a model without directly sharing their raw data and with any meaningful degradation in performance. Compared with centralized training, this architecture reduces the need to transfer or centrally store sensitive patient records and allows institutions to retain control over their local datasets. Furthermore, collaboration through FL can provide access to larger and more diverse data distributions than would typically be available to any single institution, potentially resulting in models that are more accurate and robust across heterogeneous populations. These advantages have contributed to the widespread characterization of FL as a privacy-preserving approach to Machine Learning. However, this claim can be misleading if such decentralization is paired with a complete privacy protection. Although FL eliminates the need to centralize patient information, it does not fully eliminate the possibility of information leakage. Model updates are computed directly from local data and may therefore encode information about the datasets that generated them. An adversary with access to these updates may be able to infer properties of the underlying datasets or, in some cases, recover information about individual records or again about the institution that produced a particular update. According to Yoo et al., FL represents *"a structural solution for existing data privacy violation problems of machine learning methods"* in healthcare.

Nevertheless, this form of "protection by design" may often be insufficient in a collaborative learning environment. While FL is a rapidly evolving technology with significant potential for healthcare and biomedical research, its privacy risks and broader ethical implications remain, indeed, insufficiently explored in many practical deployments. This gap between FL's privacy promise and its practical implementation constitutes the motivation for this work. Rather than assuming that the decentralized architecture of FL is inherently sufficient to guarantee patient privacy, we investigate whether and where privacy bottlenecks may arise within the original FL pipeline. Our research therefore focuses on the following questions: which privacy vulnerabilities can arise throughout a standard Federated Learning pipeline and how can these weaknesses be effectively mitigated through appropriate technical solutions. By identifying these vulnerabilities and examining suitable mitigation strategies, this work aims to provide a more critical assessment of the extent to which FL can genuinely support privacy-preserving Machine Learning.

6.2 Privacy under the GDPR and the AI Act

6.2.1 Patient Consent and Autonomy

Even if FL systems were fully privacy-preserving — which, as established above, are not — this would not by itself discharge developers or healthcare institutions from their obligations to ensure that patient data are processed on an appropriate legal and ethical basis. Patients hold a justified sense of ownership over their data and the absence of direct third parties access does not eliminate their interests in how, why and for what purposes their personal data are used. Depending on the applicable legal framework and the specific processing context, consent may therefore be required for hospitals to contribute to a federated consortium and for developers to use those data to train medical FL models, regardless of whether the raw data are ever transmitted to external parties. This requirement can nevertheless create a tension between individual control and the collective benefits of biomedical research. Strict consent procedures may indeed reduce the amount and diversity of data available for training, potentially affecting model performance and, ultimately, the precision of medical applications that FL is intended to support. Evidence suggests that this trade-off is not merely theoretical: when informed about the costs associated with strict consent requirements, some patients report a willingness to accept reduced control over their data in exchange for researches that produce public benefits and remains broadly accessible (Street et al., 2014; Richter et al., 2019; Ballantyne et al., 2022). Some bioethicists go further and argue that patients may have a prima facie obligation to participate in health research, since themselves will benefit from the generated medical knowledge (Porsdam Mann et al., 2016; Ballantyne and Schaefer, 2018). Under this view, contributing to health research may be considered a reciprocal responsibility, particularly when the resulting outcomes are expected to benefit the community (Ballantyne and Schaefer, 2020). These arguments, however, do not eliminate the importance of patient autonomy and informed decision-making. The distinction between privacy guaranteed at the architectural level and genuine patient autonomy is illustrated by the medical Machine Learning deployment described by Khetpal et al. (2022). In their case, James, a patient with a body dysmorphic disorder, discovers that his health data were used to train a model that rated rhinoplasty patients' attractiveness on a ten-point scale. He strongly objected to the system scope because he believed that it reinforced the beauty standards that had contributed to his condition. When he contacted his doctor, an administrator explained that his consent was unnecessary, since the model had been trained via FL and none of his data was ever transmitted to a third party. James nevertheless felt exploited because his data had contributed to a system fundamentally opposed to his believes and values. James' case exposes criticism towards equating data separation with ethical privacy: a system can prevent direct data sharing by design while still be deployed in ways that are ethically indefensible.

Although FL avoids direct exchange of data, participating institutions remain jointly involved in determining how the federation operates and have responsibilities concerning the purposes and means of data processing. Establishing a FL consortium may consequently require formal data protection, processing and sharing agreements defining the responsibilities of participating parties towards one another, patients and external parties, as well as the conditions governing data and model update exchange (Ballhausen et al.). The fact that no institution can directly access another institution's dataset does not necessarily mean that the exchanged model updates are anonymous or non-identifiable, as demonstrated empirically in our experiments. Consortium members should therefore avoid treating model updates as inherently privacy-preserving and should instead establish explicit governance and appropriate security measures to mitigate the risk of leakage and make partners and patients more comfortable in contributing to a federated study. Additional safeguards, such as limiting the information exchanged during training and adopting secure computation techniques, can consequently serve as trust-building mechanisms between participating institutions and patients. Consortium members must also establish clear rules concerning intellectual property, particularly in bioinformatics research where trained models may have downstream commercial applications, including patents and associated financial outcomes. Third parties requiring access to the federated infrastructure may similarly be subjected to the established data-processing agreements and security requirements.

6.2.2 Connections with the GDPR

Privacy implications of Federated Learning are closely connected to the principles established by the General Data Protection Regulation (GDPR). However, the decentralized architecture of FL should not be interpreted as being, by itself, sufficient to ensure compliance with the Regulation, but rather this depends on what information is processed, whether individuals can be identified or their characteristics inferred, and if appropriate technical and organizational safeguards are implemented.

A first point of connection concerns the scope of personal data. Under *Article 4*, personal data are broadly defined as information relating to an identified or identifiable natural person. Consequently, the fact that FL does not require conventional patient records to be transmitted to a central server does not determine that the information exchanged during training falls outside the scope of data-protection obligations. Indeed, if any information contained in an update can reasonably be linked to an identifiable person, such processing may still raise GDPR concerns. The Source Identification and Property Inference attacks investigated in this work demonstrate that the possibility of deriving information from model updates is concrete and thus must be seriously addressed when assessing the privacy implications of a FL system. This distinction is also consistent with the GDPR's treatment of pseudonymised data. Pseudonymisation does not necessarily remove information from the scope of the Regulation when individuals remain identifiable through the use of additional information. Similarly, the absence of directly identifiable patient records in a FL system should not be confused with the absence of privacy risks. The relevant question is not only whether raw data are transferred, but also whether information about individuals or their characteristics can be derived from the available information exchanged during training.

A second important connection concerns the principles governing the processing of personal data. *Article 3* applies the GDPR processing of EU data subjects even when the entity processing those data is established outside the European Union, making its territorial scope particularly relevant to international biomedical FL consortia distributed worldwide. *Article 5* establishes the principles of lawfulness, fairness and transparency, purpose limitation and data minimization, while *Article 6* establishes the conditions under which processing remains lawful. Keeping patient data within the institution can contribute to data minimization, by reducing the need to transfer or centralize sensitive records. However, this architectural property does not address the possibility that model updates themselves leak information. This issue becomes particularly relevant to the GDPR principle of data protection by design and by default established by *Article 25*. The principle requires controllers to implement appropriate technical and organizational measures designed to enforce data protection and protect the rights of data subjects. *Article 7* establishes that information about how data are processed must be sufficiently transparent to allow individuals to understand the implications of participating in a federated study and the associated risks. Similarly, *Article 32* requires the adoption of security measures appropriate to the risks linked to processing. In this context, keeping patient data local can constitute one privacy-preserving measure, but it should be complemented by mechanisms capable of reducing the information that can be inferred from model updates. Privacy protection should therefore be considered throughout the entire FL pipeline, rather than treated as an additional component to be introduced only after deployment.

Finally, the GDPR's requirements concerning transparency and data-subject rights provide a broader perspective on the risks investigated in this work. *Articles 13 and 14* establish obligations towards data subjects, *Article 15* provides a right of access to personal data, while *Article 29* establishes that, in case of automated inference, data subjects have the right to access both their original data and the final or intermediate conclusions inferred from them. *Articles 17 and 21* further establish patients rights related to erasure and objection at any time, subject to the conditions and limitations specified by the Regulation. In a federated setting, exercising such rights may raise additional technical concerns once client's local data have contributed to an aggregated global model, since removing their influence may require mechanisms beyond simply deleting the original local records. From this perspective, we are not interested in determining whether Federated Learning is in theory inherently compliant or non-compliant with the GDPR, but rather in examining whether the privacy principles and requirements established by this Regulation are sufficient to address and mitigate the types of information leakage that may arise during a collaborative training framework.

6.2.3 The AI Act and International Considerations

Beyond the GDPR, the European Union Artificial Intelligence Act (AI Act) provides an additional framework relevant to regulate the development and deployment of AI systems in healthcare. The classification of a healthcare AI system under the AI Act depends primarily on its intended purpose and on the specific use case. When Federated Learning is used to engineer AI systems falling within the high-risk categories defined by the regulation, additional requirements apply, concerning risk management, data and data governance, technical documentation, record keeping, transparency, human oversight, accuracy, robustness, and cybersecurity throughout the system's lifecycle.

The AI Act is particularly relevant to the setting considered in this work, because Federated Learning may involve multiple independent institutions, each with different datasets, technical infrastructures, and organizational practices. While this distributed structure can facilitate cross-site collaboration and improve the representativeness of the resulting model, it can also introduce heterogeneous risks across participating institutions. In particular, the privacy risks investigated in this work show that keeping patient data local does not fully eliminate the possibility of extracting sensitive information from the federated training process. This is relevant to the broader risk-management perspective required for a responsible deployment of AI systems in healthcare, where technical safeguards and organizational measures must be considered together. The regulatory context becomes more complex in international collaborations. Unlike the EU's comprehensive data-protection framework, the United States relies on a combination of federal and state legislation, including sector-specific regulations such as the HIPAA for protected health information. The legal requirements applicable to a federated deployment may therefore differ depending on the institutions involved and the jurisdictions in which they operate. For this reason, the privacy risks analyzed in this work should be intended as one component of a broader governance and risk-assessment process. Technical privacy mechanisms and explicit evaluation of inference risks are therefore necessary to ensure that the privacy benefits associated with a Federated Learning framework are achieved not only at the architectural level, but also in practice.

6.3 Source Inference Attack (SIA)

Building on the privacy risks outlined in the previous sections, this work investigates a concrete and comparatively underexplored threat within the Federated Learning landscape: whether an adversary with access to exchanged model updates can infer which institution produced a given update. This question is particularly relevant in cross-silo FL, where each client represents an identifiable institution and where the statistical characteristics of local datasets may differ substantially across participants. Positioning this threat within the broader family of inference attacks helps to clarify the specific privacy objective addressed here. Indeed, several forms of inference attack may arise in a federated scenario:

- **Membership Inference Attack (MIA):** an adversary attempts to determine whether a specific data point was included in the training data, potentially exposing the participation of an individual in a clinical study;
- **Gradient-Based MIA:** a more advanced variant of membership inference that exploits gradients or other intermediate training signals, to infer whether specific samples contributed to model training;

- **Label Inference Attack (LIA):** an adversary attempts to infer sensitive labels associated with training samples from model behavior, gradients or other information exchanged during training.

The first attack investigated in this work belongs to a fourth, related category: the Source Inference Attack (SIA), introduced by Hu et al. At a high level, SIA shifts the inference target from the individual record (Membership Inference Attack) to the client responsible for a model contribution. Indeed, rather than investigating whether a particular data point was used during a specific training round, the attacker examines which participant produced a given update. In a cross-silo setting such as the one considered here, where each client corresponds to a different institution, successful source attribution can itself constitute a privacy violation, even without the subsequent re-identification of individual patients. The sensitivity and severity of this information, indeed, arises especially from the possibility that institutional identity can be combined with additional external knowledge about the corresponding hospital, the patient population it serves, the clinical specialization adopted or data collection practices. For example, if an adversary can reliably associate an update with a hospital known to assist a particular population ethnicity or to be specialized with a specific technology, the attribution may provide additional context for interpreting the few information encoded in the updates. Therefore, a privacy risk lies not only in the information contained in an individual update, but also in the possibility of linking that information to the institution that generated it, reducing the degree of anonymity that participants reasonably expect from a federated architecture.

The SIA formulation adopted in this work differs from the one proposed by Hu et al., in which a target record is associated with the client whose local model produces the lowest loss on that record. Instead, each client's contribution is represented through statistical features extracted from its model update and source attribution is formulated as a standard supervised classification problem. The objective is to determine whether the statistical characteristics of an exchanged update contain sufficient information to distinguish its originating institution from the other participating in the consortium. Previous works provide evidence that federated updates can actually encode sufficient information about the data generating the corresponding local models. Melis et al., for example, demonstrated that an adversary can infer properties of each client's training data from the gradients produced during federated training, even without direct access to the underlying dataset. At the same time, the presence of exploitable information in the gradients does not imply that source attribution is inevitable: its success depends on the degree of statistical separation between clients, the information retained by the exchanged updates, the attacker's knowledge and the training configuration. Differential privacy has been widely proposed as a countermeasure against several forms of information leakage of this type. However, in high-dimensional domains such as genomics, privacy protection may introduce a non-negligible utility-privacy trade-off. Triastcyn and Faltings (2020), for instance, show that stronger privacy guarantees can substantially affect model utility under certain training configurations. This trade-off is particularly relevant to the present study, because the experimental analysis evaluates whether privacy-preserving mechanisms can reduce source attribution without negatively impacting the final model utility.

6.3.1 Honest-but-Curious Server: a Realistic Adversary

A Federated Learning environment involving multiple decentralized clients is considered, each holding a local subset of sensitive genomic data and a central server responsible for aggregating model updates. The adversary is assumed to be an authorized participant in the federation, rather than an external party intercepting network communication. In this way, the attacker directly exploits vulnerabilities intrinsic in the federated training pipeline, which naturally exposes information useful to complete a source inference threat. The adversary is assumed to have the following capabilities:

- **Client-Level Access:** the attacker is assumed to be an authorized participant with visibility equivalent to that of the aggregating server, either a malicious client colluding with it, or the honest-but-curious server itself, and thus has access to model updates exchanged during training;
- **Gradient Visibility:** the attacker can observe both the local updates submitted by every participating client, as well as the global model parameters distributed at each communication round;
- **Model Knowledge:** the attacker knows the architecture and relevant hyperparameters of the shared model (white-box setting);

- **No Access to Other Clients' Data:** the adversary cannot directly access the datasets held by other participating institutions, in line with FL principles;
- **Cross-Round Observation:** the attacker can record model updates, predictions, loss values and other observable quantities across communication rounds and use them to construct statistical features for future inference.

We also assume that the central server does not intentionally deploy additional privacy-preserving mechanisms unless explicitly specified in the experimental configuration. Communication between clients and the server is assumed to be protected against external interception. The relevant adversary is therefore an insider who is legitimately authorized to participate in the consortium but attempts to extract information beyond what is necessary for the intended collaborative training task. This setting is particularly relevant to cross-silo healthcare collaborations, where participating institutions may simultaneously be trusted collaborators, which aim to benefit from the final federated model, and potential malicious sources, wanting to exploit the work of others. Indeed, the privacy risks do not arise merely from the fact that the server can observe client updates, as this is an unavoidable and legitimate consequence of the FedAvg protocol itself. Rather, the risk originates because nothing in the standard aggregation protocol prevents the server from exploiting the information contained in those updates for purposes beyond computing the prescribed weighted average. The server may therefore correctly execute the aggregation process, without corrupting the global model or deviating from the training procedure, while simultaneously observing, storing and analyzing the individual contributions it legitimately receives. This corresponds to the honest-but-curious server model formalized by Hu et al., who showed that such a server can perform a source inference attack without violating the rules of a federated learning protocol.

This threat model is not merely theoretical: in real-world federated learning consortia, the server may not necessarily be an independent or neutral entity, but it may be operated by one of the participating institutions or by an external provider responsible for coordinating the process. In either case, the server has the technical ability to access and analyze individual updates and may have interests that extend beyond the aggregation task itself. Beyond source inference, previous work has also demonstrated that an honest-but-curious server can extract other forms of client-private information. For example, Dai et al. show that a server can infer the class distribution of individual clients without deviating from the prescribed protocol. Therefore, our attacks target two related forms of sensitive information. The first concerns institutional attribution: determining which hospital produced a given update can expose information about the population or clinical context represented by that contribution. When combined with external knowledge about the corresponding institution, such attribution may enable an adversary to draw inferences that would not be possible from anonymized updates alone. The second is a Property Inference Attack, which concerns the statistical characteristics of genomic data encoded in local model updates. Genomic features may correlate with sensitive attributes such as ethnic origin, ancestry, disease susceptibility or other phenotypic characteristics, meaning that even partial leakage can have significant privacy implications.

6.3.2 Scope and Security Context

The privacy analysis presented in our experiments focuses specifically on information leakage from model updates. The Source Inference Attack and Property Inference Attack investigated therefore address a different security objective from other popular attacks that, instead, aim to manipulate or disrupt the federated training process. While inference attacks seek to extract information about participating clients or their underlying data, attacks such as data poisoning, model poisoning, gradient manipulation and Byzantine attacks aim primarily to influence the behaviour or performance of the resulting global model. This distinction is particularly relevant to the threat model considered in this work, where the objective is therefore not to compromise the training process or modify the global model, but to exploit information that is already available through the participation in the consortium. In this setting, protecting communication channels alone would not prevent the attacks, since the relevant information is accessed by an adversary who is already authorized to participate in the federation and thus to observe the corresponding model updates.

Nevertheless, other security threats remain relevant in a practical federated healthcare deployment: external attackers may attempt to compromise communication channels or gain unauthorized access to the federated infrastructure. Addressing these threats would require additional mechanisms, such as client authentication, access control, encrypted communication, update validation, anomaly detection and audit logging. These broader security mechanisms were not implemented or evaluated in the present work because they address different attack objectives from the privacy risks considered in the experimental analysis. Other attacks and security mechanisms should therefore be considered complementary components of a broader security architecture for real-world federated medical AI systems and can be examined for future works on the topic.

6.4 Property Inference Attack

Beyond identifying the source client responsible for a given update, an adversary may also attempt to infer statistical properties of each client's local dataset without necessarily determining its precise identity. This type of threat is commonly referred to as a Property Inference Attack. Previous work by Melis et al. has shown that gradients and model updates can encode information about the distribution of the data used for local training, even when individual records are not directly exposed. In the federated genomic setting considered in this work, two types of dataset properties are of particular interest: the size of each client's local dataset and its cell-type composition. The attack relies on the same update-derived statistics used for client identification, namely the per-layer mean, standard deviation, ℓ_2 norm, minimum and maximum values. However, rather than predicting the identity of the client that generated an update, the inference task is formulated as a regression problem, with the objective of predicting quantitative properties of the underlying local dataset.

The first target is local dataset size, measured as the number of cells available at each client side. Although this information may appear relatively innocuous, a successful inference could reveal the scale of an institution's contribution to the federation and, potentially, provide additional information about the institution itself. The second target is cell-type composition. This choice is motivated by the source inference results, where clients with similar cell-type distributions were found to be more frequently confused by the attacker. This observation suggests that the biological composition of local datasets may influence the information encoded in federated updates and thus leave a detectable signature. To investigate this possibility more directly, the attack is applied to the cell types exhibiting the highest variability across clients, which represent the most distinctive components of the local data distributions.

6.5 Differential Privacy as Mitigation

The inference attacks investigated in our experiments are possible because model updates, although not containing raw data explicitly, remain statistically dependent on the information used to compute them. This dependency is fundamental in a Federated Learning setting: model updates must retain information about local training data in order to contribute meaningfully to the optimization of the global model but, at the same time, such information can potentially be exploited by an adversary to infer properties of the underlying dataset or of the institution that generated the corresponding update. This tension motivates the use of Differential Privacy (DP) as the principal mitigation strategy, which is one of the most widely adopted techniques for data protection in Federated Learning. Differential Privacy is not an encryption mechanism but a statistical framework for limiting the information that an output can reveal about any individual contribution to a computation. In this way, the extent to which an observer can confidently infer some attributes from the released output significantly decreases. In practice, Differential Privacy is implemented by first bounding the influence of individual contributions and then adding calibrated random noise. In Federated Learning, these operations can be applied at different levels of the training process, depending on the threat model and on which party is assumed to be trusted. The resulting privacy guarantee is governed by the amount of noise injected, the clipping bound considered and the number of training rounds. Increasing the noise generally strengthens privacy but reduces the information available for learning, producing an inherent privacy-utility trade-off.

We investigate a client-side noise-injection strategy in which each client clips its local update and perturbs it with Gaussian noise before the transmission to the server. This design avoids sending an unperturbed local update to the aggregator and is therefore appropriate for the insider threat model considered above, where an honest-but-curious server attempts to infer information from the updates exchanged during training. Indeed, since the adversary is precisely the party that would otherwise receive unperturbed updates, protecting privacy solely through server-side noise injection would offer no defence against it. Perturbing each contribution at the source before it ever reaches the server, instead, ensures that no raw update is ever exposed to the malicious entity, rather than relying on a trusted central aggregator to protect updates only after collection. Protection comes, however, with an utility cost: because each client independently perturbs its own contribution, noise accumulates across participants and communication rounds, rather than being introduced only after aggregation. Two complementary mechanisms for noise-injection are considered in our experiments. First, gradient clipping bounds the norm of each client's update before transmission. Clipping limits the influence of unusually large updates and establishes a sensitivity bound that can be used to calibrate subsequent noise injection, but it still does not prevent an adversary from exploiting information contained in the bounded update. Second, Gaussian noise injection adds random perturbations to the clipped update before leaving the client. This approach substantially reduces the information available to the attacker and thus its effectiveness, but the effect on model utility becomes increasingly severe as the noise magnitude increases.

6.5.1 Alternative Privacy-Preserving Techniques

The privacy–utility trade-off observed in our experiments is not unique to the Differential Privacy approach. Other privacy-enhancing technologies may provide different mechanisms for protecting federated model updates, potentially reducing the information exposed to an adversary without relying exclusively on noise injection. Homomorphic Encryption (HE), for example, allows computations to be performed directly on encrypted values, enabling the server to aggregate client contributions without accessing the corresponding plaintext updates. In principle, this could reduce the exposure of the client-specific signals exploited by the Source Inference Attack investigated in this work. Its main limitation, however, lies in the substantial computational and communication overhead associated with encrypted operations, particularly when applied to large neural networks. Secure Multi-Party Computation (SMPC) provides another valuable alternative, by allowing multiple parties to jointly compute a function over their inputs without directly revealing those inputs to one another. In a federated setting, SMPC can therefore prevent an individual participant or an honest-but-curious aggregator from directly accessing another client's unprotected update. However, as with HE, this protection may introduce additional computational and communication costs affecting scalability. A further alternative is represented by multi-server architectures, designed to reduce the utility degradation associated with client-side noise injection. In such approaches, clients may distribute different components of a perturbation across separate servers, allowing the aggregated model to benefit from noise removal while preventing any individual server from observing an unperturbed client contribution.

Overall, these approaches highlight that the privacy–utility trade-off observed in our experiments is not necessarily limited to a choice between applying strong noise to the entire model and accepting weaker protection to preserve utility. Alternative methods to Differential Privacy may provide different balances between privacy protection, computational cost and model performance. However, these approaches were not implemented in the present work because they require additional computational resources, more complex infrastructures or stronger trust assumptions between participating parties. A systematic comparison of these approaches is beyond the scope of our work, but represents a natural direction for future research in privacy-preserving federated genomic learning.

7 Fairness Analysis

Fairness across participating institutions is an important consideration in federated learning, particularly when the local datasets are heterogeneous in size and composition. In the considered single-cell RNA-sequencing setting, each client represents a different sequencing technology and contains a different number of cells. As a consequence, the standard Federated Averaging (FedAvg) algorithm does not give every client the same influence on the global model. While weighting clients according to their number of training examples is appropriate when optimizing the average performance over all observations, it may lead to an imbalance in which smaller institutions have considerably less influence on the global model. The fairness analysis was therefore designed around two main questions. First, we investigated whether clients containing fewer cells obtain worse model performance under standard FedAvg and whether alternative aggregation strategies can reduce the resulting differences. Second, we investigated whether these fairness properties are maintained when clients are not available in every communication round, including a scenario in which smaller clients are systematically less available.

7.1 Fairness in AI

As artificial intelligence systems are increasingly used in decision-making and other socially relevant domains, ensuring that they do not produce unfair or discriminatory outcomes has become an important research objective. Bias can arise at different stages of the AI and machine learning pipeline, including data collection, algorithm design, and human interaction with the system. In particular, machine learning models may learn and reproduce patterns of bias present in their training data, especially when the data are incomplete or unrepresentative of the population they are intended to serve. Consequently, addressing fairness requires consideration not only of model outputs, but also of the data, algorithms, and processes involved in developing and deploying AI systems.

Fairness in AI is a complex and multifaceted concept, and different definitions may be appropriate depending on the context and stakeholders involved. Among the main notions discussed in the literature are group fairness, which seeks to ensure that different groups receive equal or proportionate treatment; individual fairness, which aims to ensure that similar individuals are treated similarly; and counterfactual fairness, which considers whether a system would reach the same decision for an individual under a hypothetical change in their group membership or attributes. These notions are not necessarily mutually compatible, and achieving fairness may require balancing different objectives. In particular, efforts to reduce bias can involve trade-offs with predictive accuracy or other performance measures, meaning that improving fairness cannot always be achieved independently of other model objectives [2].

7.1.1 Fairness in Federated Learning

These considerations become particularly relevant in Federated Learning (FL), where models are trained collaboratively across multiple clients without requiring their raw data to be centralized. By keeping data on local devices and sharing model updates instead, FL can improve data privacy while enabling learning from distributed data sources. However, the decentralized nature of FL also introduces substantial heterogeneity among clients, including differences in data distributions, computational capabilities, network conditions, and participation frequency. Such heterogeneity can lead to unequal contributions to the training process, as some clients may be selected more frequently or may have a greater influence on the resulting global model than others.

In this context, fairness in FL is not limited to treating all clients identically. Rather, it concerns the extent to which different clients have a meaningful opportunity to contribute to the training process and obtain adequate performance from the resulting global model. Client selection is particularly important in this regard: strategies that prioritize clients with better computational resources, faster communication, or higher local performance may repeatedly favor the same participants and underrepresent resource-constrained clients. Similarly, data imbalance, heterogeneous client capabilities, and aggregation mechanisms can contribute to disparities in model performance and utility across clients. Several approaches have therefore been proposed to improve fairness in federated learning, including modifications to client selection, aggregation weights, resource allocation, and the optimization objective. q-Fair Federated Learning is an example of an aggregation-based approach that assigns greater importance to clients experiencing higher losses, thereby attempting to reduce disparities in model performance across clients. Such approaches, however, generally involve a trade-off between fairness and global model utility. Increasing the importance of poorly performing clients may improve the consistency of performance across the federation while simultaneously reducing the average performance of the global model or affecting convergence [7].

7.1.2 Partial Participation in Federated Learning

An additional challenge in federated learning arises from the fact that clients cannot always participate in every communication round. Although full participation may be assumed in some theoretical settings, practical federated learning systems are typically characterized by intermittent client availability. Clients may differ substantially in their computational resources, battery capacity, network connectivity, and communication bandwidth, which can prevent some of them from participating in particular rounds. Moreover, involving all clients simultaneously may become impractical in large-scale federated systems because of communication and computational costs.

As a result, only a subset of clients may contribute to the global model in a given communication round. Partial participation can affect the optimization and convergence of federated learning, particularly when client data are heterogeneous or non-IID. From a fairness perspective, the participation pattern is also important because clients that participate less frequently have fewer opportunities to contribute their local updates to the global model. If participation is not uniform across clients, differences in availability can therefore become an additional source of disparity in the federated training process.

This issue becomes particularly relevant when participation is correlated with client characteristics. For example, smaller or resource-constrained institutions may be less likely to participate regularly than larger institutions. In such a setting, clients with fewer local observations may already have less influence under number-of-examples-weighted aggregation, while lower participation further reduces their opportunities to contribute to the global model. Partial participation can therefore amplify existing disparities rather than simply introducing an independent source of variability.

7.1.3 Motivation for the Fairness Analysis

These considerations motivate the fairness analysis performed in this work. Rather than evaluating the federated model only according to its aggregate performance, we explicitly examine the performance obtained by each institution. In particular, we investigate whether clients with fewer single-cell observations are disadvantaged by the standard FedAvg aggregation rule and whether alternative aggregation strategies can reduce such disparities. We further investigate whether these effects persist under partial client participation, including a scenario in which smaller clients are systematically less likely to participate.

7.2 Experimental Setting

The analysis was performed on seven clients corresponding to the sequencing technologies *celseq*, *fluidigmc1*, *inDrop1–4*, and *smarter*. The number of local training cells ranges from 510 to 2,884. This heterogeneity is relevant to the fairness analysis because the three smallest clients also correspond to non-droplet or full-length sequencing technologies, whereas the remaining clients are droplet-based. Consequently, client size and sequencing technology are not independently controlled in the experimental partition. This represents an important limitation when interpreting a relationship between the number of cells and model performance.

7.3 Aggregation Strategies

To investigate whether changing this weighting can improve fairness, we compared standard FedAvg with three alternative approaches. All aggregation strategies were evaluated under the same client partition, model architecture, local training procedure, number of federated rounds, and optimization settings. Therefore, differences between the resulting models can be attributed primarily to the aggregation strategy.

7.3.1 Standard Federated Averaging

Standard Federated Averaging was used as the baseline aggregation strategy. At the end of each communication round, the server aggregates the locally updated parameters received from the participating clients. If $\boldsymbol{\theta}_k$ denotes the parameters obtained from client k and n_k is the number of local training cells, the standard FedAvg update is

$$\sum_{k=1}^{K} \frac{n_k}{\sum_{j=1}^{K} n_j} \boldsymbol{\theta}_k^{(t+1)}. \tag{1}$$

The corresponding aggregation weight of client k is therefore

$$\frac{n_k}{\sum_{j=1}^{K} n_j}. \tag{2}$$

This weighting scheme means that a client with a larger number of cells contributes proportionally more to the global model. From the perspective of global empirical risk minimization, this is a natural choice because it makes the global objective approximately equivalent to the loss over the union of all local observations. However, from a fairness perspective, it also means that a small client can have considerably less influence than a large client.

This motivated the comparison with aggregation strategies that reduce the dependence on client size or explicitly prioritize clients with poorer performance.

7.3.2 Uniform Weighting

Each client contributes equally to the global aggregation, independently of its number of training cells. This means that, K clients participate in a communication round, the aggregation becomes

$$\frac{1}{K} \sum_{k=1}^{K} \boldsymbol{\theta}_k^{(t+1)}, \tag{3}$$

with

$$w_k^{\text{uniform}} = \frac{1}{K}. \tag{4}$$

This provides a simple baseline for assessing whether removing the size-based weighting of FedAvg benefits smaller clients. It is important to note that uniform weighting does not explicitly consider how well each client is performing. A small client and a large client receive equal weight even if one of them has substantially higher validation loss. Therefore, while uniform weighting can improve representation fairness, it does not directly target worst-performing clients.

7.3.3 q-Fair Federated Learning

Unlike uniform weighting, q-FFL [5] explicitly gives greater importance to clients with higher losses. The strength of this fairness pressure is controlled by q. In our implementation, an unnormalized aggregation weight is first computed for each participating client as

$$\tilde{w}_k = n_k (F_k + \epsilon)^q, \tag{5}$$

where n_k is the number of local training cells, F_k is the client's cached training loss, and ϵ is a small constant for numerical stability. The weights are then normalized during aggregation, resulting in the effective client weight

$$w_k = \frac{n_k (F_k + \epsilon)^q}{\sum_{j=1}^{K} n_j (F_j + \epsilon)^q}. \tag{6}$$

When the cached loss is not available, the implementation falls back to $\tilde{w}_k = n_k$. The case $q = 0$ corresponds to standard FedAvg, while increasing q increasingly favors clients with higher losses. Because the loss used to determine the current aggregation weights is only available after local evaluation, the implementation uses the client's loss from the previous federated round as a one-round-lagged estimate when computing the weights for the current round. We evaluated $q = 0.5$, 1, 2, and 5. The range allows us to investigate the trade-off between a relatively weak fairness correction and a substantially stronger one.

7.3.4 Adaptive q-FFL

An adaptive version of q-FFL was also evaluated, in which the value of q is adjusted during training based on the observed change in Jain's fairness index. This was intended to provide a self-tuning alternative to selecting a fixed fairness parameter.
The approach was inspired by the adaptive mechanism proposed in Pei [9]. In F^3, the fairness mechanism is implemented by adding a regularization term to the clients' local training objective, with the corresponding regularization weight adapted by the server according to the evolution of the fairness metric. We could not directly apply this approach because scVI encapsulates its local training procedure through PyTorch Lightning, and introducing an additional fairness term would have required substantial modifications to the scVI training process.
Instead, we applied the same adaptive principle to the q-FFL parameter q, which can be fully controlled by the server during aggregation. After each evaluation round, q is updated, for use in the aggregation of the next round, according to the relative change in Jain's fairness index:

$$\Delta_F(t) = \frac{J(t) - J(t-1)}{J(t-1)}, \tag{7}$$

where $J(t)$ denotes Jain's fairness index computed from the clients' evaluation losses at round t.
The new value of q is then computed as

$$q(t+1) = \text{clip}\left(q(t) - \alpha\Delta_F(t), q_{\min}, q_{\max}\right), \tag{8}$$

where α controls the magnitude of the adaptation and $q_{\min}$ and $q_{\max}$ define the allowed range of q.
Because a higher Jain's index corresponds to improved fairness, an increase in $J(t)$ produces a decrease in q, progressively moving the aggregation closer to FedAvg. Conversely, when Jain's index decreases, q is increased, strengthening the q-FFL reweighting of clients with higher loss. The adaptive mechanism therefore attempts to apply stronger fairness corrections only when they are needed, rather than maintaining a fixed value of q throughout training.
For reproducibility and diagnostic purposes, the value of q and the corresponding Jain's index are recorded after each communication round. The resulting trace is stored in a CSV file and can be used to inspect how the fairness objective affects the aggregation strength throughout training.

7.4 Client-level performance and fairness metrics

For each trained global model, the validation negative ELBO (Evidence Lower Bound) was measured separately on each client. In scVI, the ELBO is a measure used during training to evaluate how well the model represents the observed gene-expression data. It combines information about how well the model reconstructs the observed data with how well the learned latent representation agrees with the assumptions of the model. A higher ELBO corresponds to a better fit, but in our implementation its negative value is used as the loss. Therefore, lower negative ELBO values indicate better model performance.
The negative ELBO was calculated on the validation data of each client after federated training. This gives one performance value for each client, allowing us to compare how well the global model performs across the different institutions.
Global performance was measured using the number-of-cells-weighted average of the client-level negative ELBO values. The same weighting was used for all aggregation strategies to make the comparison fair. To measure how equally the model performs across clients, we used Jain's fairness index:

$$J(\mathbf{x}) = \frac{\left(\sum_{k=1}^{n} x_k\right)^2}{n\sum_{k=1}^{n} x_k^2}, \tag{9}$$

where x_k denotes the validation loss of client k and n is the number of clients. Higher values indicate more equal losses across clients, with $J = 1$ corresponding to equal losses. The analysis was complemented by the worst-client and best-client losses.

7.5 Accounting for sequencing depth

An additional issue emerged during the initial analysis. Since the scVI negative ELBO is calculated from the reconstruction of the observed expression counts, its magnitude is affected by the sequencing

depth of the cells. Clients with larger library sizes can therefore exhibit larger raw loss values even when the relative quality of the fitted model is not necessarily worse. We quantified the mean library size of each client and found a strong correspondence between sequencing depth and raw validation loss.
This led us to distinguish between two forms of apparent unfairness.

- *Representation-driven unfairness* refers to disparities caused by the federated optimization procedure, for example when smaller clients receive less influence under number-of-examples-weighted FedAvg.
- *Scale-driven apparent unfairness* arises when the numerical value of the loss differs systematically because of intrinsic properties of the client data, such as sequencing depth.

To investigate whether a client-size effect remains after accounting for this confounding factor, we additionally considered a diagnostic depth-adjusted loss:

$$\widetilde{L}_k = \frac{L_k}{\overline{d}_k}, \tag{10}$$

where L_k is the raw validation loss and $\overline{d}_k$ is the mean library size of client k. This normalization was not used as a training objective. Instead, it was used to determine whether the relationship between client size and model performance persisted after reducing the effect of sequencing depth.

7.6 Partial client participation

The second part of the fairness analysis considered a more realistic federated learning scenario in which not every institution is available at every communication round. This is relevant in practical deployments because institutions may temporarily be offline, have limited computational resources, or be unable to participate in a particular training round.
Three participation regimes were considered:

- *Full participation* corresponds to the baseline setting in which all clients participate in every communication round
- *Random dropout*, makes each client independently available with probability $p = 0.7$ at each round. Since the same probability is used for every client, this regime introduces temporary client unavailability without explicitly favoring large or small institutions
- *Size-correlated dropout*, introduces a systematic relationship between client size and availability. Smaller clients are assigned a lower probability of participation, while larger clients have a higher availability probability. In the implemented experiment, clients below approximately 850 cells use $p_{\text{small}} = 0.5$, whereas larger clients use $p_{\text{large}} = 0.9$. This regime is designed to test a particularly challenging form of federated unfairness: clients that are already represented by fewer cells may also have fewer opportunities to contribute updates to the global model

Because client sizes are unknown to the server in advance, the first communication round used full participation to let the server learn each client's identity and training-set size from its replies, with unlearned clients treated as fully available in the meantime and a minimum number of participants enforced each round to avoid stalling. Dropout applied only to the training phase: every client still evaluated the global model each round regardless of participation, so reported fairness and performance metrics always reflect all clients, while only each client's influence on the model itself depended on its availability.
The participation experiments were crossed with different aggregation strategies, focusing on standard FedAvg ($q = 0$) and q-FFL, including the strongest fairness-oriented setting ($q = 5$). This allows us to determine whether the fairness improvements observed under full participation remain effective when clients are intermittently unavailable.
For each participation regime, we evaluated convergence of the global validation loss, final global performance, and client-level fairness. Jain's fairness index and the variance of client losses were used to determine whether partial participation increases disparities between institutions. The observed participation frequencies were also monitored to verify that the simulated availability mechanism behaved consistently with the intended participation regime.

8 Results and Discussion

While the previous sections established the theoretical risks associated with privacy and fairness in a Federated Learning approach, here we empirically evaluate these threats under the proposed experimental setting. The remainder of this section is organized accordingly to the two complementary perspectives. The first part presents the privacy experiments including, in order, the Source Inference and Property Inference Attacks, followed by the evaluation of the proposed Differential Privacy noise-based mitigation strategy. The second part presents the fairness analysis, examining client-level performance under standard FedAvg and alternative aggregation strategies, as well as the effect of partial client participation, on convergence, global performance, and fairness. In particular, we first compare standard FedAvg, uniform weighting, q-FFL and adaptive q-FFL to investigate the trade-off between global model performance and fairness across clients. We then extend the analysis to partial client participation by evaluating FedAvg and different q-FFL configurations under full participation, random dropout, and size-correlated dropout. Where relevant, experimental findings are presented through tables and figures and discussed in relation to our privacy and fairness objectives.

8.1 Source Inference Attack Results

The Source Inference Attack, also referred to as Client Identification in the experiments, evaluates whether federated model updates contain sufficient information to identify the client that generated them. The attack is implemented as a supervised classification task, using a Random Forest classifier with 200 estimators and all the other hyperparameters fixed at their default values. The attacker is trained and evaluated using a round-based, rather than random, train-test split: updates from earlier communication rounds are used for training, while those from later, held-out rounds are reserved for the final evaluation. This choice reflects a more realistic adversary that must generalize to future communication rounds not yet observed, instead of exploiting updates from the same rounds used during training. The training process converged successfully, with both training and validation loss decreasing consistently across communication rounds (see Figure 4 - Federated training process). The attack is then evaluated using accuracy and per-client precision, recall and F1-score (Figure 5 - Client Identification Attack Classification Report). A random guessing model having an accuracy equal to 0.143, i.e. 1/7 (number of clients), is used as baseline for comparison. Results show that federated model updates contain a substantial and exploitable signal about their originating institution. Using only aggregated statistics extracted from the exchanged updates — mean, standard deviation, L2 norm, minimum and maximum weight values — the classifier correctly identifies the source client in 69.2% of the held-out cases, substantially above the random-guessing baseline of 14.3%. Performance, however, varies considerably across clients, with F1-scores ranging from 0.431 to 0.973. In particular, clients 0 and 4 achieve valuable F1-scores of 0.914 and 0.973, respectively.

The confusion matrix provides further insight into the structure of the classification errors (Figure 6 - Client Identification Attack Confusion Matrix). Misclassifications are not distributed uniformly across clients: when an update is incorrectly attributed, it is frequently confused with a specific other client. For example, client 1 is correctly identified in 11 cases, but is classified as client 6 in 8 cases. This suggests that some pairs of clients may produce statistically similar update patterns, making their contributions more difficult to distinguish. Two analyses were performed to investigate possible explanations for these differences in per-client performance. First, we assess whether institutions with fewer patients are more susceptible to re-identification. The correlation between per-client F1-score and local dataset size is found to be positive and moderate (Pearson correlation coefficient = 0.524 and Figure 7 - F1-score – Dataset Size Correlation). In this setting, clients with larger local datasets tend to be more consistently identifiable, suggesting that larger datasets may produce more stable client-specific update signatures. This differs from the intuition commonly associated with membership inference, where smaller datasets can increase the relative influence of individual samples (Bagdasaryan et al., 2019). Second, the similarity between clients' cell-type compositions is positively correlated with the number of classification errors between them (r = 0.434 and Figure 8 - Client - celltype similarity correlation)). Clients with a similar biological compositions tend to be confused more frequently by the classifier. These analyses provide a biologically grounded interpretation of the confusion matrix representation, assessing that systematic classification errors can be explained by similarities in the underlying biological data and that biological characteristics of the local dataset leave a detectable trace in the statistical signature encoded by the corresponding updates.

Each client update consists of 25 individual weight tensors, corresponding to the parameters (weights and biases) of every dense layer in the scVI architecture. The interpretability analysis further shows that client-identifying information are not distributed uniformly across these 25 tensors. SHAP analysis identifies three weight tensors, namely 6, 18 and 24 (Figure 9 - Layer Importance via SHAP), as the most informative components for the attack. Tensor 6 (namely, z_encoder.mean_encoder.bias) is associated with the mean of the latent representation, while tensor 18 (decoder.px_decoder.fc_layers.Layer1.0.bias) contributes to the decoder's reconstruction of gene expression. Tensor 24 (decoder.px_dropout_decoder.bias), instead, contributes to the estimation of technical dropout. These components may therefore capture systematic characteristics of the local data distribution or sequencing process and contribute more actively to the success of the identification attack. To assess whether access to the complete model update is actually necessary, the attack was repeated using only these three tensors. Restricting the classifier to this small subset of parameters increased the accuracy from 69.2% to 77.4%, demonstrating that effective client identification can be achieved even without requiring access to the entire updates (Figure 10 - Important layers - Classification report and Figure 11 - Important layers - Confusion matrix). This procedure can be understood as a form of feature selection: by discarding the less informative tensors, the classifier is trained on a lower-dimensional but more discriminative representation of each update, reducing the noise introduced by less relevant parameters, rather than uselessly increasing model capacity.

Overall, the source inference results demonstrate that decentralizing raw patient data does not prevent institutional re-identification and that this identifiability is concentrated in a small subset of model components, which make the identification attack even easier. These first findings motivate the subsequent Property Inference Attacks, which investigate whether the information contained in the same updates can be exploited to reconstruct properties of each client's local dataset beyond its identity.

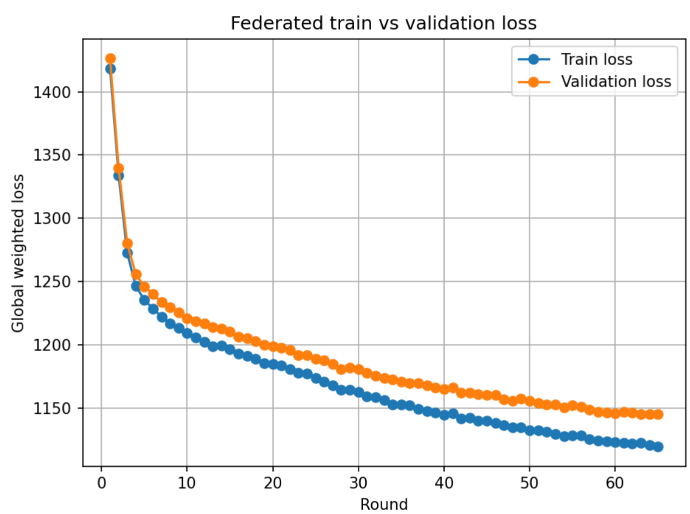

Figure 4: **Federated training progress -** federated training and validation loss over communication rounds, with early stopping applied

Client Identification Attack with Random Forest

	precision	recall	f1-score	support
0	1.0	0.842	0.914	19.0
1	0.344	0.579	0.431	19.0
2	1.0	0.316	0.48	19.0
3	1.0	0.526	0.69	19.0
4	1.0	0.947	0.973	19.0
5	0.8	0.632	0.706	19.0
6	0.528	1.0	0.691	19.0
accuracy	0.692	0.692	0.692	0.692
macro avg	0.81	0.692	0.698	133.0
weighted avg	0.81	0.692	0.698	133.0

Figure 5: **Client Identification Attack Classification Report** - classification report for the Client Identification Attack, showing precision, recall and F1-score per client, along with macro and weighted averages, evaluated on held-out communication rounds

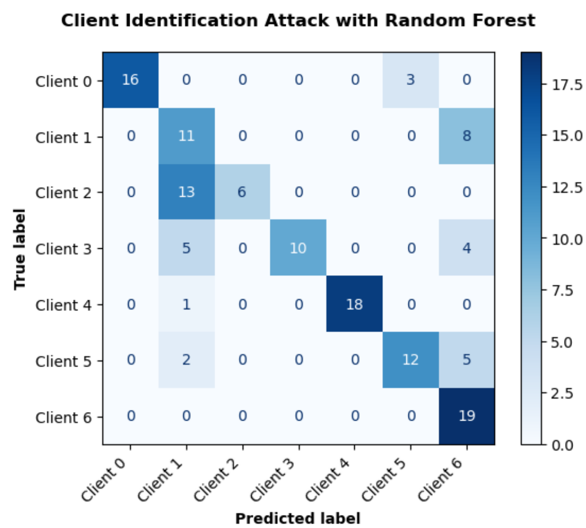

Figure 6: **Client Identification Attack Confusion Matrix** - confusion matrix comparing true and predicted client identities on held-out rounds. Off-diagonal entries indicate misclassified updates, revealing which institutions are most frequently confused with one another

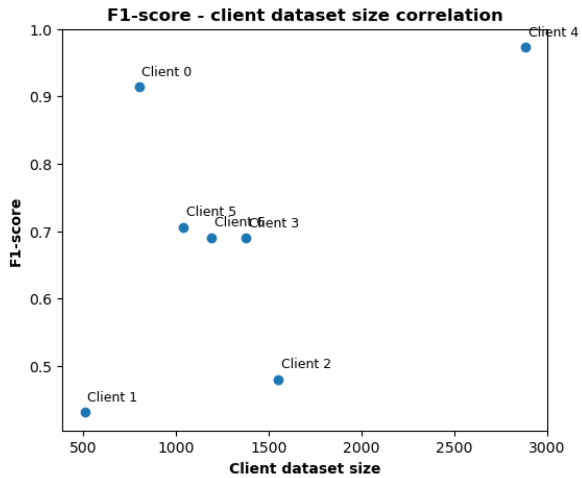

Figure 7: **F1-score – Dataset Size Correlation** - per-client F1-score plotted against local dataset size, testing whether re-identification risk varies systematically in relation with institution size

Classification errors and celltype similarity

Client A	Client B	Celltype similarity	Total errors	Rank
1	6	0.944	8	1
3	6	0.925	4	2
1	3	0.903	5	3
2	5	0.896	0	4
1	5	0.892	2	5
3	5	0.871	0	6
1	2	0.862	13	7
1	4	0.855	1	8
3	4	0.852	0	9
0	4	0.849	0	10
4	6	0.821	0	11
0	5	0.770	3	12
4	5	0.769	0	13
5	6	0.757	5	14
0	3	0.750	0	15
2	4	0.710	0	16
2	3	0.681	0	17
2	6	0.666	0	18
0	1	0.622	0	19
0	2	0.578	0	20
0	6	0.537	0	21

Figure 8: **Client - celltype similarity correlation** - relationship between cell-type composition similarity (cosine similarity) and the total number of classification errors for each client pair, showing that clients with more similar cell-type distributions are more frequently confused by the attacker

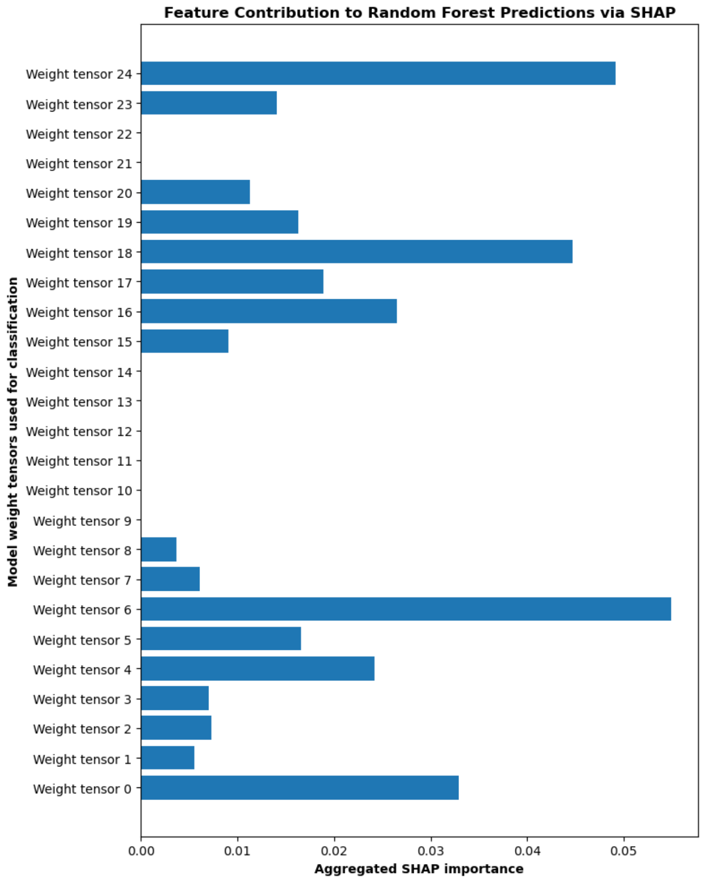

Figure 9: **Layer Importance via SHAP** - aggregated SHAP feature importance per model update, identifying which components of the model contribute most to the attack's success

Client Identification Attack on layers [6, 18, 24]

	precision	recall	f1-score	support
0	1.0	1.0	1.0	19.0
1	0.938	0.789	0.857	19.0
2	0.826	1.0	0.905	19.0
3	0.692	0.474	0.562	19.0
4	0.475	1.0	0.644	19.0
5	1.0	0.316	0.48	19.0
6	1.0	0.842	0.914	19.0
accuracy	0.774	0.774	0.774	0.774
macro avg	0.847	0.774	0.766	133.0
weighted avg	0.847	0.774	0.766	133.0

Figure 10: **Important layers - Classification report** - classification report for the Client Identification Attack restricted to the three most important weight tensors (6, 18, 24) identified via SHAP, evaluated on the same held-out rounds as the full-model attack

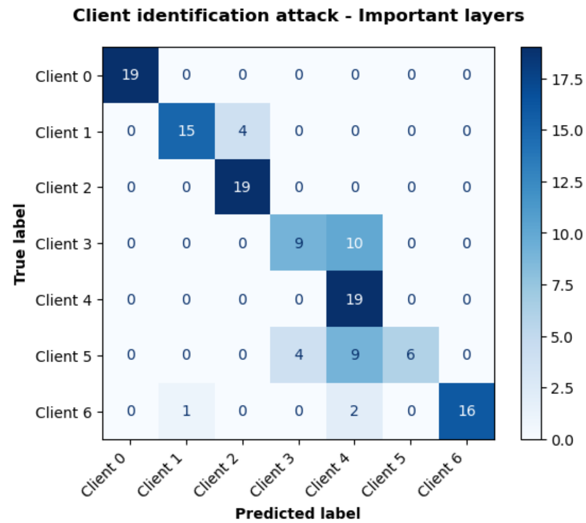

Figure 11: **Important layers - Confusion matrix** - confusion matrix for the Client Identification Attack restricted to the three most informative weight tensors (6, 18, 24), for comparison against the previous full-model results

8.2 Property Inference Attack Results

Having shown that client identity can be successfully inferred from federated model updates, we next investigate whether the same updates can also reveal statistical properties of each client's local dataset. We focus on two types of properties: local dataset size and cell-type composition. This second attack is formulated as a regression problem, using a Random Forest regressor having 200 estimators and the same round-based train/test split used for client identification. Performance is evaluated through Mean Absolute Error (MAE) and the coefficient of determination (R^2), with lower MAE and higher R^2 indicating a more accurate reconstruction. MAE measures the average absolute difference between predicted and true values, expressed in the same unit as the target variable, while R^2 measures the proportion of variance in the target variable explained by the model. A value of 1 indicates a perfect prediction, 0 indicates performance no better than always predicting the target's mean and negative values communicate a performance worse than this trivial baseline. R^2 is scale-independent, allowing direct comparison across properties having very different scales.

Results show that local dataset size can be inferred with an R^2 of 0.687 and a MAE of 314.9 cells, relative to a mean dataset size of 1337.4 cells - computed across all clients and rounds in the held-out test set. Therefore, predictions explain a substantial proportion of the variation in dataset size across clients; their distribution across held-out rounds also shows a moderate spread within individual clients, indicating that the signal exploited by the regressor remains stable across rounds (Figure 12 - Dataset size reconstruction). The attack was then extended to cell-type composition to determine whether biological characteristics of local datasets are also encoded in model updates. Among the four most variable cell types, ductal cells are the most accurately inferred, achieving an R^2 of 0.693 and a MAE of 0.043. Alpha cells achieve an R^2 of 0.547 and a MAE of 0.0697, while beta and acinar cells achieve lower but still positive R^2 values of 0.434 and 0.371, respectively (see Table 1). These results indicate that the possibility of infer biological properties varies across cell types, but that multiple aspects of local cell-type composition leave detectable signals in the exchanged updates.

The property inference results therefore extend the privacy risk beyond client re-identification. An adversary does not necessarily need to determine which institution produced an update in order to extract sensitive information from it. The ability to reconstruct dataset size and cell-type composition demonstrates that model updates can reveal characteristics of the underlying data distribution itself. These results can be further combined with those of the Source Inference Attack to reconstruct a broader client profile (Figure 13 - Reconstructed Client Profile). By combining client identification with predictions of dataset size and cell-type composition, an adversary can potentially obtain information about both who produced an update and what type of data contributed to it. For most clients, the combined predictions reconstruct the main characteristics of the underlying client and local dataset, illustrating how multiple inference attacks can provide a comprehensive and informative picture of the entire consortium. Taken together, these findings demonstrate that the privacy risks associated to federated model updates are not limited to institutional attribution. Statistical and biological properties of local datasets can also be inferred from the same information available to an attacker, highlighting the need for mitigation strategies capable of reducing information leakage while preserving the utility of the federated model.

Property	**MAE**	$\mathbf{R^2}$
Dataset size	314.9	0.687
Cell type: alpha	0.0697	0.547
Cell type: ductal	0.0432	0.693
Cell type: beta	0.0673	0.434
Cell type: acinar	0.0490	0.371

Table 1: **Summary of regression performance** (MAE, R^2) across the property inference attacks evaluated in this work: dataset size and the four most variable cell types across clients. Ductal cell proportion and dataset size are the most reliably inferable properties, both achieving R^2 of approximately 0.68

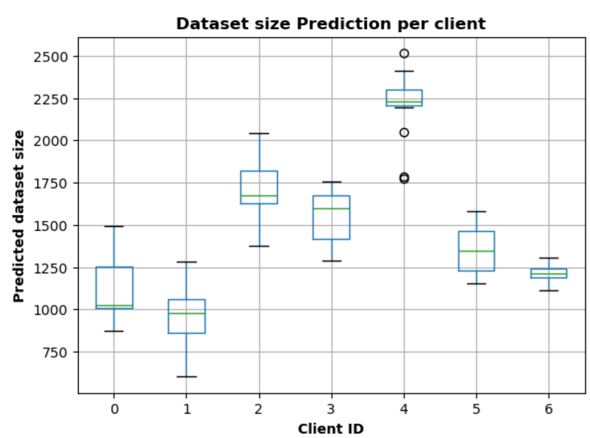

Figure 12: **Dataset size reconstruction** - distribution of predicted local dataset sizes per client, aggregated across held-out test rounds. The spread within each client's box reflects prediction variability across rounds

Reconstructed Client Profile

Client	True Tech	Predicted Tech	True Size	Predicted Size	True CellType Prop	Predicted CellType Prop
0	0	0	803	1124	0.324	0.243
1	1	1	510	968	0.043	0.088
2	2	1	1550	1702	0.06	0.098
3	3	3	1379	1543	0.174	0.145
4	4	4	2884	2219	0.103	0.114
5	5	5	1042	1370	0.223	0.189
6	6	6	1194	1214	0.0	0.062

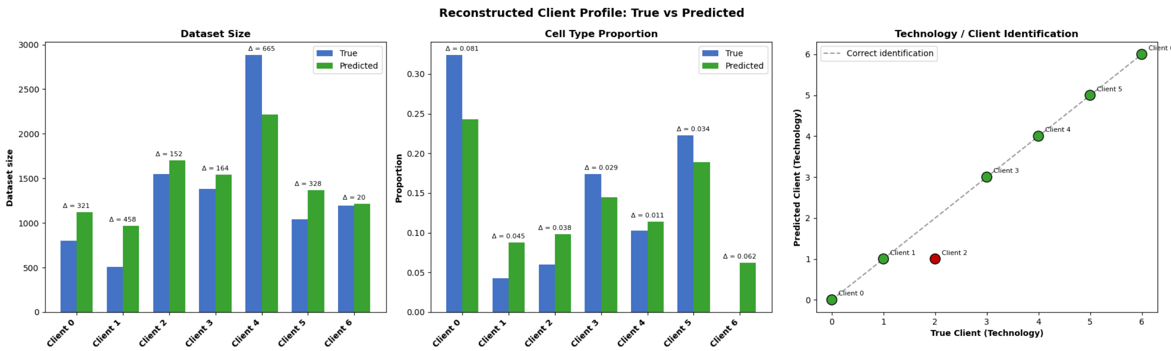

Figure 13: **Reconstructed Client Profile** - summary of the reconstructed client profile obtained by combining the three attacks: client/technology identification, dataset size inference and cell-type proportion inference. For each client, predicted values are averaged across all test-round updates attributed to that client and compared against ground truths. A visual comparison of the true and predicted values across the three reconstructed attributes is also reported: local dataset size (left), dominant cell-type proportion (center), and client/technology identity (right). Annotated values in the first two panels indicate the prediction error absolute value; points in the rightmost panel indicate correct (green) versus incorrect (red) client identity

8.3 Differential Privacy Mitigation Results

This section evaluates whether Local Differential Privacy can mitigate the privacy vulnerabilities identified in the previous experiments and how this protection affects model utility. Two mitigation strategies are compared: untargeted noise injection, in which noise is applied to all model parameters, and targeted noise injection, in which perturbation is restricted to the three model components identified by the SHAP analysis as the most informative for client identification. For each strategy, we experiment with three increasing levels of noise (0.3, 0.6 and 1.0), evaluating client-identification accuracy as an indicator of attack effectiveness and the final validation loss as a measure of model utility.

At noise level 0, both attack accuracy and validation loss correspond to the original, unperturbed federated model. As noise is introduced, the two strategies exhibit markedly different privacy–utility trade-offs. Under untargeted Differential Privacy, increasing the noise substantially reduces the attacker's ability to identify the source client (Figure 14 - Privacy vs Utility Trade-off Untargeted Differential Privacy). At noise multipliers of 0.3, 0.6 and 1.0, attack accuracy decreases from 0.692 (obtained when noise multiplier = 0) to 0.095, 0.114, 0.071 respectively, falling even below the random-guessing baseline of 0.143. This indicates that perturbing the entire model update before sending it to the server effectively suppresses the client-specific information exploited by the attacker. However, this privacy improvement comes at a substantial cost to model utility. The final validation loss increases considerably as the noise level grows, exceeding an order of magnitude relative to the no-noise baseline at the highest noise level: 1150 at noise level 0 vs 50000 when noise multiplier is set to 1. Untargeted DP therefore provides strong protection against source inference, but substantially compromises the performance of the federated model. Figure 15 provides a more detailed view of the attack performance when the highest tested noise multiplier of 1 is introduced. Compared with the unmitigated setting, the classification report shows a substantial reduction in the ability to correctly identify the source client. The corresponding confusion matrix further illustrates how the structure of the attacker's errors changes under Differential Privacy. In the original setting, when a client was misclassified, it was often confused predominantly with a specific alternative client, reflecting similarities between client data distributions. Under a noise multiplier of 1, however, misclassification errors become both more frequent and more broadly distributed across all clients. When the attacker fails to identify the correct source, the corresponding update can be confused with multiple different clients rather than consistently with a single one. This suggests that the injected noise is working properly and does not simply increases confusion between previously similar clients, but more generally disrupts the overall structure exploited by the classifier for the source identification attack. The targeted Differential Privacy strategy produces, instead, a different trade-off (Figure 16 - Privacy vs Utility Trade-off Targeted Differential Privacy). Here, noise is applied only to the three tensors identified as the most informative for client identification, namely tensors 6, 18 and 24, substantially reducing the impact of perturbation on the overall model utility. Across the tested noise levels, the final validation loss remains close to the no-noise baseline of 1150, reaching approximately 1175 at the highest noise multiplier of 1. At the same time, attack accuracy decreases to 0.333, 0.238 and 0.262 at noise multipliers of 0.3, 0.6 and 1.0, respectively. Although these values represent a substantial reduction in attack effectiveness compared to the unmitigated value of 0.692 when no noise is injected, they remain above the random-guessing baseline. Interestingly, increasing the noise beyond 0.3 does not produce a proportional improvement in privacy under the targeted strategy suggesting that, once the most informative components have been sufficiently perturbed, additional noise provides diminishing privacy benefits. In contrast, the substantial preservation of performance shows that targeting the components responsible for the strongest privacy leakage can avoid much of the utility degradation observed with uniform perturbation.

Therefore, the comparison between these two tested strategies highlights a clear privacy–utility trade-off. Untargeted Differential Privacy provides the strongest protection against client identification, but at the cost of a substantial degradation in model utility. Targeted DP, on the other hand, achieves weaker privacy protection capabilities, but preserves the performance of the federated model considerably better. These findings suggest that privacy protection does not necessarily require to perturb uniformly the entire updates: identifying and protecting the components that carry the strongest privacy-sensitive signal may provide a more practical compromise between privacy and utility. More broadly, the targeted alternative can be interpreted not as a replacement for stronger global privacy mechanisms, but as a more selective mitigation strategy particularly suitable for scenarios where preserving model utility is an important requirement, such as in clinical applications and decision-making.

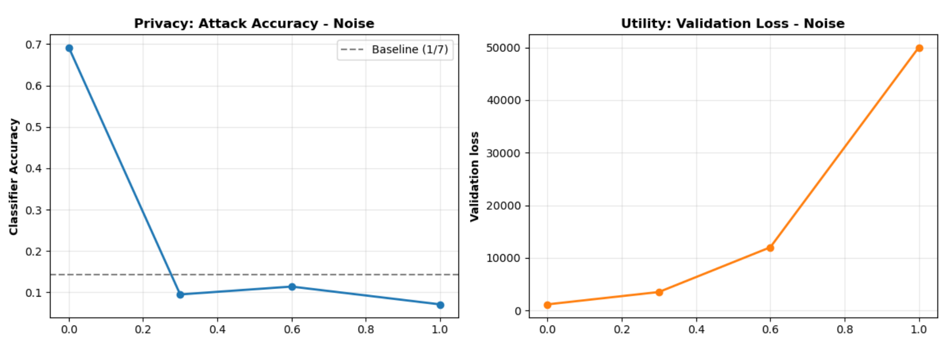

Figure 14: **Privacy vs Utility Trade-off Untargeted Differential Privacy** - privacy-utility trade-off under untargeted Local Differential Privacy. Left: client identification attack accuracy as noise multiplier increases, compared against the random-guessing baseline (dashed line). Right: final validation loss of the federated model at the same noise levels

Client Identification Attack with Random Forest

	precision	recall	f1-score	support
0	0.0	0.0	0.0	6.0
1	0.0	0.0	0.0	6.0
2	0.286	0.333	0.308	6.0
3	0.0	0.0	0.0	6.0
4	0.0	0.0	0.0	6.0
5	0.0	0.0	0.0	6.0
6	0.125	0.167	0.143	6.0
accuracy	0.071	0.071	0.071	0.071
macro avg	0.059	0.071	0.064	42.0
weighted avg	0.059	0.071	0.064	42.0

Figure 15: **Classification Report and Confusion Matrix under noise multiplier = 1**

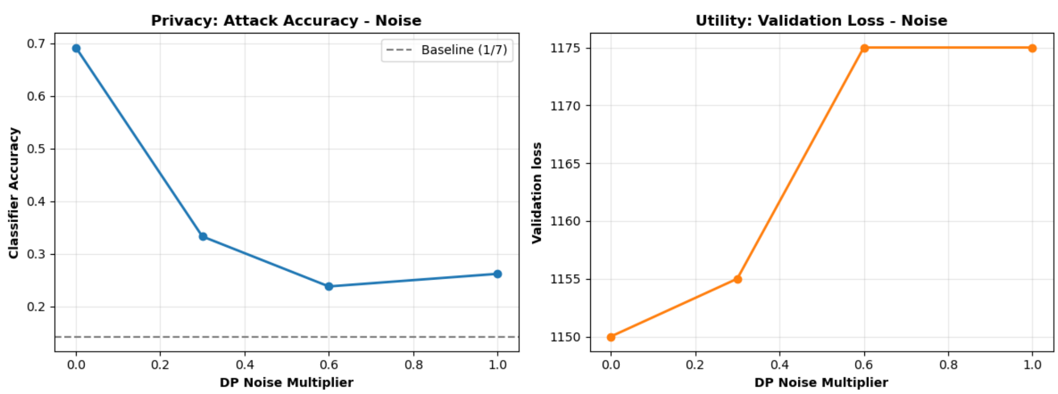

Figure 16: **Privacy vs Utility Trade-off Targeted Differential Privacy** - Privacy-utility trade-off under targeted Local Differential Privacy, with noise injected only into the three most informative weight tensors (6, 18, 24). Left: client identification attack accuracy as a function of the noise multiplier, compared against the random-guessing baseline (dashed line). Right: final validation loss of the federated model at the same noise levels

8.4 Fairness across clients under full participation

We first evaluated whether the number of cells available at each client was associated with poorer model performance under standard FedAvg. The initial analysis of the raw validation negative ELBO showed a negative association between client size and final loss (Spearman $r = -0.786$, $p = 0.036$). Taken at face value, this result would suggest that clients with fewer cells tend to obtain worse (higher) performance. However, inspection of the individual clients indicated that client size alone could not explain the observed differences. In particular, client 6 had a number of cells comparable to other clients but exhibited a substantially larger validation loss.

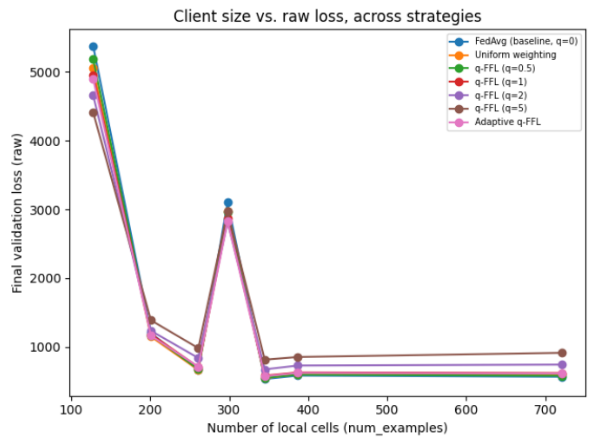

Figure 17: Comparison of final validation loss vs. client size across seven aggregation strategies (FedAvg baseline, uniform weighting, q-FFL with q=0.5, 1, 2, 5, and adaptive q-FFL)

We therefore investigated mean library size as a possible confounding factor. The resulting client-level comparison showed a strong correspondence between sequencing depth and raw validation loss. Client 1 had the highest mean library size, approximately 10,962 counts per cell, and also the highest validation loss. Client 6 had the second-highest mean library size, approximately 10,910 counts per cell, consistently with its unexpectedly high loss. Conversely, clients 5, 3, and 2 had lower mean library sizes of approximately 8,246, 7,187, and 7,506 counts per cell, respectively, and also exhibited lower validation losses.

client_id	**n_cells**	**mean_lib_size**	**valid_loss**
0	201	10268.771484	1235.16
1	128	10961.942383	4659.88
2	387	7506.309570	729.71
3	345	7187.620117	673.57
4	721	7196.999512	743.07
5	261	8246.619141	838.61
6	298	10910.443359	2813.80

Table 2: Client statistics.

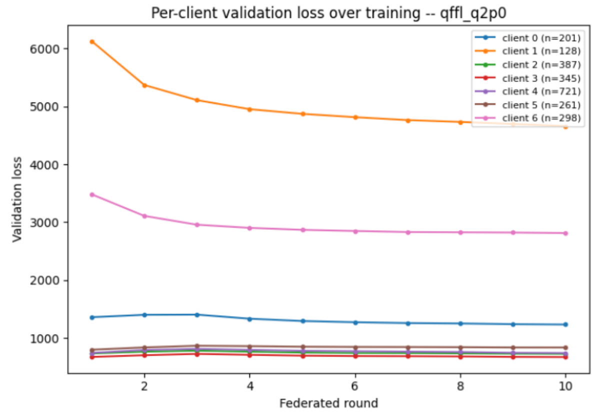

Figure 18: Validation loss trajectories for all 7 clients over 10 federated rounds under q-FFL (q=2)

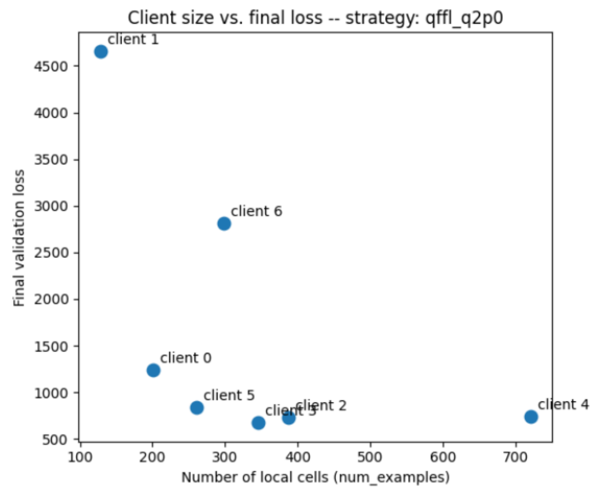

Figure 19: Final validation loss by client size under q-FFL (q=2)

This pattern indicates that a substantial component of the apparent client-level unfairness is related to sequencing depth rather than to the number of training cells alone. This conclusion is further supported by the correlation analysis reported in Table 3.

Table 3: Correlation between client characteristics and validation loss. The first column shows the correlation between mean library size and raw loss. The second shows the correlation between client size and depth-adjusted loss.

Strategy	Library size vs. raw loss	Client size vs. adjusted loss
FedAvg ($q = 0$)	$r = 0.814,\ p = 0.026$	$r = -0.534,\ p = 0.217$
q-FFL ($q = 2$)	$r = 0.815,\ p = 0.025$	$r = -0.516,\ p = 0.236$
q-FFL ($q = 5$)	$r = 0.835,\ p = 0.019$	$r = -0.494,\ p = 0.260$

The correlation between mean library size and raw validation loss remains strong and statistically significant across the tested aggregation strategies, with correlation coefficients between 0.814 and 0.835. Moreover, the relationship is essentially unchanged when increasing the q-FFL parameter. This is expected because changing the aggregation rule cannot remove the dependence of the scVI loss scale on the number of counts that must be reconstructed.
After applying the diagnostic depth normalization, the correlation between client size and validation loss became substantially weaker. The correlation remained negative across the tested strategies, rang-

ing from approximately −0.49 to −0.53, but none of the corresponding p-values was below 0.2. Therefore, the results do not provide statistically significant evidence of a client-size effect after accounting for sequencing depth. Nevertheless, the consistent negative direction of the estimated correlations suggests that a residual size-related tendency may remain.
These results should be interpreted cautiously because only seven clients are available. With such a small number of observations, statistical tests have limited power and individual clients can strongly affect the estimated correlations. Consequently, the results cannot establish the absence of a size-related fairness problem. Instead, they indicate that sequencing depth is a major confounding factor that must be considered when comparing client-level scVI losses.

8.5 Effect of aggregation strategy on client fairness

Although sequencing depth explains a substantial part of the differences in raw loss, changing the aggregation strategy can still modify how equally the resulting global model performs across clients. Table 4 summarizes the final Jain fairness indices obtained under the different aggregation strategies.

Table 4: Final-round Jain fairness index for the tested aggregation strategies.

Strategy	Jain (raw loss)	Jain (depth-adjusted loss)
FedAvg ($q = 0$)	0.500	0.575
Uniform weighting	0.517	0.598
q-FFL ($q = 2$)	0.585	0.683
q-FFL ($q = 5$)	0.651	0.758

For the raw loss, standard FedAvg obtained a Jain index of 0.500. Uniform weighting produced a modest improvement to 0.517. In contrast, q-FFL produced substantially larger improvements as the fairness parameter increased, reaching 0.585 for $q = 2$ and 0.651 for $q = 5$. A similar pattern was observed after depth adjustment: the Jain index increased from 0.575 under FedAvg to 0.598 with uniform weighting, 0.683 with $q = 2$, and 0.758 with $q = 5$.
The monotonic increase in Jain's index with q is consistent with the intended mechanism of q-FFL. Since clients with higher losses receive greater influence in the aggregation, the global model is encouraged to improve the performance of the clients that are currently performing worst. Importantly, the same trend is observed for the depth-adjusted loss. Therefore, the improvement in fairness cannot be attributed solely to the normalization of sequencing-depth differences. Instead, q-FFL appears to actively reduce disparities in client-level model performance.
Uniform aggregation also improves fairness compared with standard FedAvg, but the effect is substantially smaller. This suggests that simply removing the dependence of aggregation weights on the number of local cells is less effective than explicitly targeting high-loss clients.

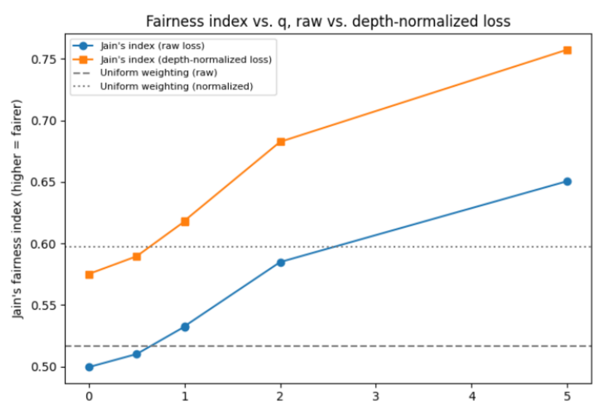

Figure 20: Jain's fairness index as a function of the q-FFL exponent, computed on both raw and depth-normalized loss. Fairness improves monotonically with increasing q for both metrics, with depth-normalized loss showing consistently higher fairness scores than raw loss

8.6 Trade-off between global performance and fairness

The fairness improvements obtained through q-FFL must be considered together with their effect on overall model performance. Standard FedAvg optimizes the global objective using the number of local examples as the aggregation weight, and therefore naturally prioritizes performance on clients with larger datasets. In contrast, q-FFL deliberately changes this objective by increasing the influence of clients with higher losses.

The resulting behaviour illustrates the expected fairness–performance trade-off. Increasing q produces increasingly equal client-level performance, as shown by the increase in Jain's index, but the stronger fairness pressure can reduce the performance of the global model. In particular, the $q = 5$ configuration provides the strongest fairness improvement but also exhibits slower convergence and a higher final global loss than the lower-q configurations. Thus, q should not be interpreted as a parameter for maximizing fairness independently of model utility. Instead, it controls the balance between optimizing average performance and protecting clients that perform poorly.

This trade-off is particularly relevant in federated bioinformatics. A globally optimal model according to a cell-weighted objective may provide good performance on the largest institutions while performing relatively poorly on smaller or systematically different datasets. Conversely, an aggregation strategy that gives excessive priority to the worst-performing clients may improve equality while sacrificing overall predictive quality. The appropriate choice therefore depends on whether the application prioritizes average performance, robustness across institutions, or a balance between the two.

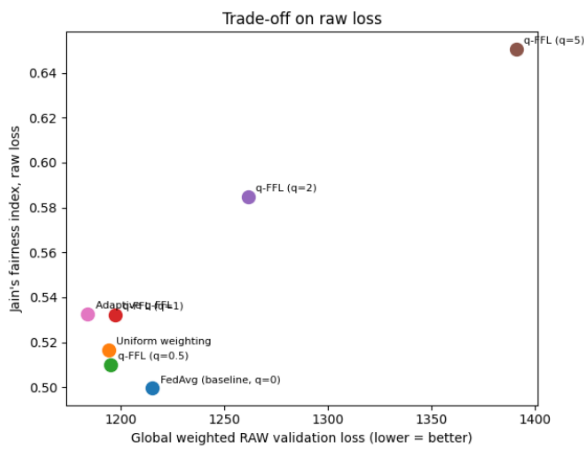

Figure 21

8.7 Effect of partial client participation

We next evaluated whether the fairness behaviour observed under full participation is preserved when clients may be unavailable during training. The full-participation setting provides the reference convergence behaviour.

Figure 22 shows the evolution of the global number-of-cells-weighted validation loss over the ten communication rounds. Under full participation, the lower-q configurations converge smoothly, with $q = 0$ reaching a final loss of approximately 1210 and $q = 2$ reaching approximately 1240. In contrast, $q = 5$ remains at a substantially higher loss, around 1355 at the end of training. This is consistent with the fairness–performance trade-off observed in the previous experiments: increasing the fairness pressure improves equality across clients but can reduce the overall performance of the global model.

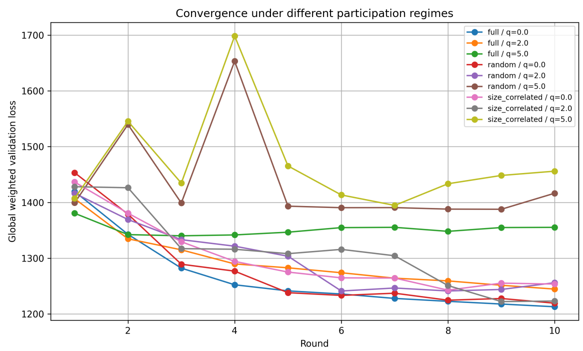

Figure 22: Global number-of-cells-weighted validation loss over federated training under full participation, random dropout, and size-correlated dropout. Each participation regime is evaluated with $q = 0$, $q = 2$, and $q = 5$.

Under random dropout, FedAvg and $q = 2$ remain relatively close to the corresponding full-participation curves. The final losses are approximately 1220 and 1255 respectively. Thus, random temporary unavailability does not appear to substantially impair convergence for these aggregation strategies over the considered training period.

However, a pronounced instability is observed for $q = 5$ under random dropout. In particular, the convergence curve exhibits a large spike around round 4, reaching approximately 1650, before decreasing again and ending at around 1415. This behaviour is consistent with an interaction between partial participation and the extreme sensitivity of $q = 5$ weighting. Because dropout is applied only to the training phase, every client evaluates the global model every round, so the cached loss used for q-FFL weighting is refreshed for all clients each round and does not become stale for non-participating clients. Instead, the instability arises because random dropout changes which subset of clients is available for training each round, while the L_k^q term with $q = 5$ strongly amplifies even ordinary, non-stale differences in loss between clients. If a client with a comparatively higher loss happens to be selected for training while competing high-loss clients are unavailable, it can temporarily receive disproportionate aggregation weight, pulling the global model sharply toward its local optimum. This transient dominance is corrected in subsequent rounds, either as the global model adjusts and that client's loss decreases, or as a different subset of clients is drawn. This provides a plausible mechanistic explanation for the observed instability rather than treating the spike as random noise.

The size-correlated dropout regime produced shows a more heterogeneous convergence pattern. Under FedAvg, the global loss decreases from approximately 1435 to 1255 by the final round. The final value is slightly higher than the full-participation FedAvg result, although the difference is relatively small. For $q = 2$, the loss decreases to approximately 1220, which is close to the final performance of full-participation FedAvg and lower than the corresponding full-participation $q = 2$ result. The strongest instability is again observed for $q = 5$. In this case, the loss rises to approximately 1700 at round 4 and remains substantially higher than the other configurations, ending at approximately 1455. Thus, size-correlated dropout combined with strong q-FFL weighting produces the least favourable convergence behaviour among the configurations considered.

Overall, the convergence results suggest that moderate q-FFL weighting ($q = 2$) is relatively robust to partial participation, whereas the combination of high q and client dropout can lead to substantial instabilit, regardless of whether availability is random or size-correlated. These observations should, however, be interpreted cautiously because the experiment covers only ten communication rounds and uses a single participation realization.

8.7.1 Per-client performance under partial participation

The final-round client-level losses provide additional insight into the observed global behaviour. Figure 23 reports the validation loss for each client and each combination of participation regime and aggregation strategy.

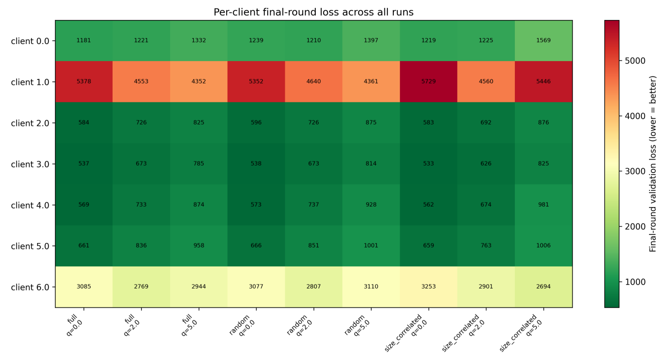

Figure 23: Final-round validation loss for each client across the participation regimes and aggregation strategies. Lower values correspond to better performance

The heatmap reveals a pronounced heterogeneity between clients that is consistent across the different experimental configurations. In particular, client 1 exhibits substantially larger losses than all other clients, with final losses ranging approximately from 4350 to 5730. Client 6 also consistently has a higher loss than clients 0, 2, 3, 4, and 5, with values between approximately 2700 and 3250. In comparison, clients 2–5 generally remain below approximately 1000.
This pattern is important for the interpretation of the fairness results. The same clients tend to remain difficult to fit across participation regimes and aggregation strategies. Therefore, client-level disparities cannot be explained solely by the temporary availability of institutions. The persistence of these differences is consistent with the earlier analysis showing that intrinsic properties of the client datasets, including sequencing depth, strongly affect the magnitude of the scVI negative ELBO. At the same time, the aggregation strategy changes the magnitude of these differences.

8.7.2 Fairness–performance trade-off under partial participation

The relationship between global performance and client-level fairness is summarized in Figure 24. The horizontal axis shows the final global weighted validation loss, where lower values indicate better overall performance, while the vertical axis shows Jain's fairness index, where higher values indicate greater equality across clients.

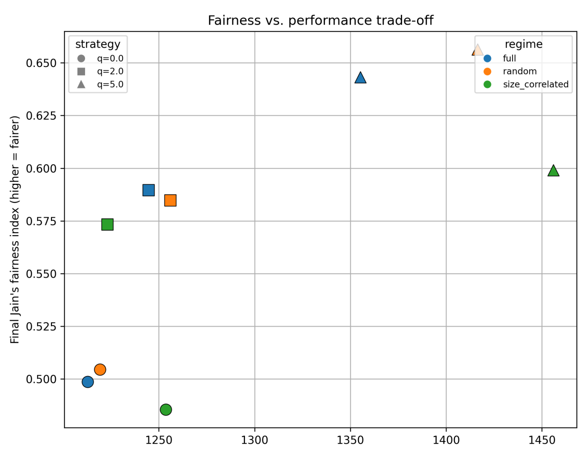

Figure 24: Trade-off between final global performance and client-level fairness under the different participation regimes. Circle, square, and triangle markers correspond to $q = 0$, $q = 2$, and $q = 5$, respectively

In general, the figure shows a clear progression as the q-FFL parameter increases. Under full participation, increasing q produces the expected trade-off between global performance and fairness. Moving from FedAvg to increasingly strong q-FFL weighting results in progressively higher Jain's fairness index, but also in higher global validation loss. This provides the reference behaviour against which the two partial-participation regimes can be compared.

Random dropout largely preserves this behaviour. The three configurations follow a similar progression to the full-participation setting, with stronger q-FFL weighting producing higher fairness at the cost of global performance. This indicates that randomly removing clients from individual rounds does not fundamentally alter the fairness benefit of q-FFL. The main difference is observed for the strongest setting, $q = 5$, for which the global performance is worse than under full participation. This is consistent with the convergence instability observed previously and suggests that high-q q-FFL is more sensitive to temporary client unavailability.

The size-correlated regime presents a more challenging setting. Under FedAvg, fairness is slightly lower than in the other two regimes, which is consistent with the reduced participation of the smallest client. Interestingly, moving to $q = 2$ produces a substantial improvement in fairness while maintaining relatively good global performance. In this case, moderate q-FFL therefore provides a favourable compromise between the two objectives. Increasing the fairness pressure further to $q = 5$ results in an additional fairness improvement, but with a much larger degradation in global performance. Moreover, the fairness achieved by $q = 5$ under size-correlated participation remains below that obtained under full or random participation with the same value of q. This suggests that loss-based reweighting alone cannot completely compensate for systematic differences in client availability. When smaller clients have fewer opportunities to participate, the aggregation strategy must compensate not only for differences in client losses but also for differences in how frequently their updates reach the server.

Overall, the comparison highlights the importance of considering client availability together with the aggregation strategy when evaluating fairness. Random dropout has a relatively limited effect on the fairness–performance trade-off, whereas size-correlated dropout makes the trade-off more pronounced. Among the configurations tested, moderate q-FFL ($q = 2$) provides a particularly favourable compromise in the size-correlated setting, while the more aggressive $q = 5$ configuration achieves higher fairness at a substantially larger performance cost. The corresponding final performance and fairness values are reported in Table 5.

Table 5: Final global performance and client-level fairness under the different participation regimes. Values are reported for $q = 0$, $q = 2$, and $q = 5$. Lower global validation loss indicates better overall performance, whereas higher Jain's index indicates greater fairness.

Participation regime	Strategy	Global loss	Jain index
Full	$q = 0$	≈ 1213	≈ 0.499
Full	$q = 2$	≈ 1245	≈ 0.590
Full	$q = 5$	≈ 1355	≈ 0.643
Random	$q = 0$	≈ 1219	≈ 0.505
Random	$q = 2$	≈ 1256	≈ 0.585
Random	$q = 5$	≈ 1416	≈ 0.665
Size-correlated	$q = 0$	≈ 1253	≈ 0.488
Size-correlated	$q = 2$	≈ 1218	≈ 0.573
Size-correlated	$q = 5$	≈ 1455	≈ 0.598

Turning to participation itself, Figure 25 provides a sanity check that the two availability mechanisms behaved approximately as intended. Under random dropout, the empirical participation rates remain in a relatively narrow range across the seven clients and show no clear dependence on client size. This is consistent with the use of a common availability probability for all clients.

The size-correlated regime produces a different pattern. The two smallest clients have a substantially lower empirical participation rate than the remaining clients, while the larger clients participate in most of the communication rounds. This is consistent with the threshold-based availability mechanism used in the experiments: clients below the 850-cell threshold are assigned a lower availability probability ($p_{\text{small}} = 0.5$), whereas larger clients use $p_{\text{large}} = 0.9$. The empirical rates do not reproduce these probabilities exactly because the experiment contains only a limited number of rounds, but they clearly show the intended difference in availability between the smallest clients and the larger ones.

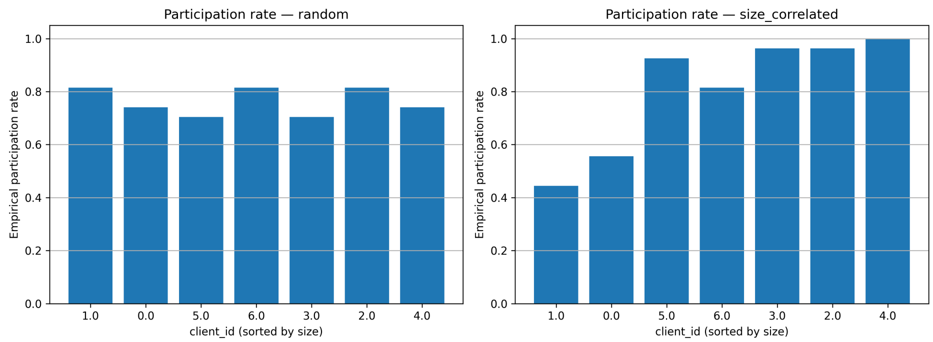

Figure 25: Empirical per-client participation rates under random dropout (left) and size-correlated dropout (right), with clients sorted by local training set size

This participation imbalance is important when interpreting the fairness results. Under size-correlated dropout, smaller clients not only have fewer cells and therefore less representation in standard FedAvg, but they also have fewer opportunities to contribute updates to the global model. The availability experiment therefore introduces an additional source of unequal representation and allows us to evaluate whether fairness-oriented aggregation can compensate for it.

9 Conclusions and future directions

Federated Learning represents a promising approach for biomedical and bioinformatics research, as it enables multiple institutions to collaboratively train models without requiring the centralization of their local datasets. This is particularly relevant for single-cell and genomic applications, where datasets are tipically distributed across institutions and contain sensitive information. However, the results of this work highlight that decentralizing the data does not, by itself, guarantee either privacy or fairness. This project investigated these two aspects from complementary perspectives. From a privacy perspective, we evaluated whether both information about the participating institutions and their local datasets could be inferred from federated model updates, and investigated differential privacy as a possible mitigation strategy. From a fairness perspective, we examined whether differences in client size translated into differences in model performance, compared alternative aggregation strategies designed to improve fairness, and investigated how partial client participation affects both convergence and client-level fairness. Taken together, our experiments showed that the benefits of Federated Learning must be considered together with the risks introduced by the federated training process itself. Privacy and fairness are not automatic properties of a decentralized architecture, but both require explicit evaluation and appropriate design choices. In particular, our results illustrate the trade-offs that arise when attempting to improve these properties, either through privacy-preserving mechanisms that may reduce model utility or through fairness-oriented aggregation strategies that instead could affect global performance and convergence. Therefore, these findings support the use of Federated Learning as a valuable framework for collaborative biomedical research, while also emphasizing the need for a careful validation before deployment in real-world healthcare settings. Future work should extend the experimental evaluation to larger numbers of clients, repeated training runs, longer training periods and additional privacy and fairness-preserving mechanisms. Such evaluations would provide a more robust understanding of the conditions under which Federated Learning can provide useful collaborative modelling, while maintaining appropriate privacy and fairness guarantees.

Privacy conclusions - From a privacy perspective, the experimental analysis has demonstrated that Federated Learning should not be considered inherently privacy-preserving simply because raw data remain local. In the federated genomic setting considered in this work, model updates retain enough information to support both source inference and property inference attacks. While identifying the institution that produced a particular update is not necessarily harmful by itself, the associated privacy risk becomes substantially more significant when that identity can be linked to sensitive characteristics of the population served by the institution. The property inference results further demonstrate that the information exposed through model updates extends beyond client identity: characteristics such as local dataset size and cell-type composition can also be inferred, potentially allowing an adversary to reconstruct a complete, detailed profile of each participating institution. On the other hand, differential privacy experiments demonstrated that these risks can be mitigated, but also highlighted the inherent privacy-utility trade-off in doing so. Untargeted noise injection provides stronger protection against the source inference attack, reducing the attacker's performance to random levels, but at the cost of substantial degradation in model utility. In contrast, targeted noise injection preserves model utility much more effectively by perturbing only the components identified as particularly informative for client identification, although it provides only partial protection against such an attack. Results presented in this work are based on a single federated training run and should therefore be interpreted in light of the inherent stochasticity of neural network training, including factors such as weight initialization, batch ordering and dropout. Repeated runs with identical hyperparameters may produce different model updates and, consequently, different levels of attack performance. A more robust evaluation would therefore require repeating the experiments across multiple independent training runs and reporting the resulting variability in attack effectiveness and model utility. This represents an important direction for future work. Future work could also extend this privacy evaluation by implementing additional attack scenarios discussed in the introductory sections. For example, an external adversary capable of intercepting communications could be investigated, to assess what information may be exposed when model updates are observed both during their transmission from clients to the server and when the updated global model is distributed back to the participants. Consequently, future works could also focus on experimentally evaluate the alternative privacy-preserving techniques discussed in this report and compare their privacy guarantees, computational requirements and impact on model utility with those of differential privacy. Such a comparison could help to determine whether different mitigation strategies offer a more favorable privacy-utility trade-offs for federated genomic learning.

Fairness conclusions - From a fairness perspective instead, results show that the answer to the first research question is more nuanced than initially expected. Under standard FedAvg, clients with fewer cells did not consistently exhibit worse model performance once differences in sequencing depth were taken into account. In fact, the analysis showed that raw scVI loss is strongly affected by library size, making it an imperfect metric for directly comparing clients with different sequencing depths. This highlights the importance of distinguishing between differences caused by client representation and differences caused by the characteristics of the underlying data.

The comparison of aggregation strategies nevertheless showed that fairness can be improved through q-FFL. Increasing q generally resulted in higher Jain's fairness index, although this came at the cost of global model performance. Among the configurations tested, moderate q-FFL provided a more favourable fairness–performance trade-off than the most aggressive setting.

Regarding the second research question, random client unavailability did not substantially alter this behaviour, and the fairness improvement of q-FFL was largely preserved. In contrast, when smaller clients were also made less available, the fairness–performance trade-off became more challenging. Moderate q-FFL still improved fairness, while the strongest setting resulted in a substantial performance cost and did not fully recover the fairness achieved under full participation. Moreover, the combination of high q and partial participation produced convergence instabilities, likely related to the extreme sensitivity of high-q weighting to whichever subset of clients happened to be available for training in a given round, an effect that partial participation introduces by continually reshuffling that subset.

Overall, these results suggest that fairness in federated single-cell learning cannot be attributed to client size alone. Data characteristics, aggregation strategy, and client availability can all contribute to differences in client-level performance and should therefore be considered together when evaluating fairness.

A main limitation of this study is that, due to memory constraints, the experiments were limited to ten communication rounds and a single random seed. Therefore, the observed differences should be interpreted as trends rather than statistically robust estimates. Future work should repeat the experiments with multiple seeds and longer training, investigate more robust depth-independent evaluation metrics, and explore aggregation strategies that explicitly account for client availability and are more robust to reshuffling of the participating client subset across rounds.

10 Reproducibility

The code required to reproduce the experiments and inspect the corresponding results is available in the project repository at the following link:

`https://github.com/VirginiaFoschi/AI_ETHICS_Proj_Foschi_Triulzi.git`

The repository is organized into several directories, with the federated learning experiments implemented in the `exercise-scvi` directory. This directory contains the `pyproject.toml` configuration file, which defines the project dependencies and the main configuration parameters used to run the experiments.

Environment and data preparation - To reproduce the experiments, first create a folder named `data_centralized` and download the dataset from the following link into the folder:

`https://drive.switch.ch/index.php/s/J8NLl6KBVWwQEEA`

The required Conda environment can then be created directly from the supplied environment file (`fl-course-env.yaml`). This file contains the complete Conda environment specification, including the required Python packages and their dependencies. After creating the environment, it can be activated with:

```
conda activate fl-course-env
```

The script `preliminaries.py` is used to download and prepare the datasets required by the subsequent experiments:

```
python preliminaries.py
```

If necessary, navigate to the correct project folder with cd ∼/ before running these commands.

Experiment selection - The experiments implemented in `exercise-scvi` are selected through the `strategy` field in `exercise-scvi/pyproject.toml`. This field can take one of three values:

```
strategy = "privacy"
strategy = "fairness"
strategy = "availability"
```

The three values correspond respectively to the privacy experiments, the client-fairness experiments and the experiments investigating partial client participation. Additional fields in `exercise-scvi/pyproject.toml` control the specific configuration of the selected experiment.

Privacy experiments - When `strategy = "privacy"` is selected, first download the appropriate `client_updates` folder from Google Drive. The first provided link contains the updates required to reproduce the Client Identification Attack and the untargeted Differential Privacy experiments with noise multipliers of 0.3 and 0.6:

`https://drive.google.com/drive/folders/1mceP2WSYdzPZ1_d-4IFkVhOUaL7eYCZd?usp=share_link`

The second link provides the updates required for the experiments with a noise multiplier of 1 and for targeted Differential Privacy:

`https://drive.google.com/drive/folders/1Nh8gyXAbQMi1LuRLRVn_0xrcBqThbXET?usp=share_link`

After downloading the desired `client_updates` folder, copy it inside the directory named `FL`, replacing the existing folder if necessary. Do not change the `client_updates` folder name. Once the folder is in place, open the `attacks_and_mitigation.ipynb` notebook located in the `privacy_analysis` folder and simply run the cells sequentially, to reproduce the corresponding experiments and inspect the results. Alternatively, you can also generate the `client_updates` folder from scratch by selecting the `strategy = "privacy"` and rerunning the federated training. To experiment with different Differential Privacy configurations, modify the `dp_noise_multiplier` and `dp_clip_norm` in `exercise-scvi/pyproject.toml`, according to the values you wish to test. Set `dp_noise_multiplier` value to 0 if you don't want Differential Privacy to be applied. In the current version of the code, the privacy strategy is configured to apply targeted Differential Privacy, reflecting the more favorable privacy-utility trade-off observed in our experiments. To switch between untargeted and targeted noise injection, comment/uncomment the call to `add_dp_noise` or `add_dp_noise_targeted` respectively, inside the train() function of `exercise-scvi/app/client_app.py`

Fairness experiments - When `strategy = "fairness"` is selected, the aggregation strategy is specified through the `fairness_strategy_name` parameter. The available options are:

```
fairness_strategy_name = "qffl"
fairness_strategy_name = "adaptive_qffl"
fairness_strategy_name = "uniform"
fairness_strategy_name = "fedavg_ploss"
```

For q-FFL, the parameter `fairness_q` specifies the initial value of q. The parameters `fairness_q_alpha`, `fairness_q_min`, and `fairness_q_max` control the behaviour of the adaptive q-FFL variant by defining the step size, lower bound, and upper bound of q, respectively.
The fairness experiments can therefore be reproduced by selecting the desired aggregation strategy and adjusting the corresponding parameters in `pyproject.toml` before launching the experiment.
The trained models and corresponding outputs for the fairness experiments are provided in the `fairness_results` directory. Therefore, the fairness analysis notebooks can be executed directly using the provided results, without requiring the federated training process to be run again. These notebooks allow the user to investigate the results for the selected aggregation strategy and reproduce the client-level metrics, fairness measures, and visualizations presented in this report.

Partial client participation experiments - Partial client participation is selected using `strategy = "availability"`. The participation regime is then controlled by `participation_regime`, which can take the values `"full"`, `"random"`, or `"size_correlated"`.
For the random participation regime, `availability_p` specifies the probability that a client is available in each communication round. For the size-correlated regime, clients with fewer than `availability_size_threshold` cells are considered small clients and are assigned the probability `availability_p_small`, while larger clients are assigned `availability_p_large`. The configuration used in the experiments was:

```
availability_size_threshold = 850
availability_p_small = 0.5
availability_p_large = 0.9
```

The aggregation used during the availability experiments is controlled by `availability_q`. Setting `availability_q = 0.0` corresponds to FedAvg-equivalent weighting, whereas larger values introduce the q-FFL weighting mechanism. The random participation process is controlled by `availability_seed`, ensuring that the dropout sequence can be reproduced. Finally, `availability_bootstrap_rounds` specifies the number of initial rounds during which all clients participate before the selected dropout regime is activated. As for the fairness experiments, the trained models and results are already provided, in this case in the `pp_results` directory. The corresponding analysis can therefore be reproduced directly using the provided notebook (`partial_partecipation_results.ipynb`), without rerunning the federated training experiments.

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